

Credit Allocation Mix in Peer-to-Peer Lending: A Network Study on Bondora

Social Network Analysis - Group 7

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1 Introduction

With the advent of online peer-to-peer (P2P) lending platforms, the traditional methods of financial intermediation have been usurped by individual choice; with no stringent third parties involved in transactions, individuals have greater power in sourcing credit. This presents new realities for individuals disenfranchised by the traditional financial system (Alistair Milne & Paul Parboteeah, 2017). Traditional financial literature shows that individually, investors exhibit behavioural biases (Agrawal, 2012; Ayal, Hochman, & Zakay, 2011) in how they choose their investments, however, numerous studies demonstrate that these behavioural biases are present among non-professional P2P borrowers. For example, Herzenstein, Dholakia, & Andrews (2011) and Lee & Lee (2012) suggest that users cluster around popular loans, exhibiting ‘herding’ behaviour. Literature focuses predominantly on funding success (Yao, Chen, Wei, Chen, & Yang, 2019), however, Ayal, Bar-Haim, & Ofir (2018) suggests that there is insufficient attention on how borrowers form their loan portfolios. As the authors indicate, this area of research is important in understanding why less professional investors deviate from known models of rational investor behaviour. Our study aims to model and understand how P2P borrowers’ behaviours influence what kinds of loans they acquire. In subsequent discussions, we consider investors as rational participants in credit markets, as described by theory, whereas borrowers refer to the users of P2P platforms.

Investigating behavioural biases, Ayal et al. (2018) find that investors’ familiarity with specific assets can make them seem less risky, making recognisable assets more attractive. However, such assets lead to undiversified portfolios. Evidence with P2P platforms supports the familiarity bias in lenders, as Galak, Small, & Stephen (2010) demonstrate that lenders choose borrowers who are similar across demographic attributes such as gender, occupation, and ethnicity. Moreover, Amar, Ariely, Ayal, Cryder, & Rick (2011) explains that borrowers show an aversion to holding a *large number of debts*, preferring to first reduce debt count, even if they had other debts with higher interest rates. This irrational behaviour demonstrates that individual borrowers face behavioural biases on the *type* and *number* of debts they select. Based on these notions, we formulate our first study, asking:

Research Question 1: *Do borrowers vary in their loan use type and frequency based on their individual and loan characteristics?*

This question is answered using linear Quadratic Assignment Procedure (QAP) regressions, as we can represent similarities across loan uses and attributes as matrices, weighted by loan use frequency, as inputs in the model.

Using data from LendingClub, Serrano-Cinca, Gutiérrez-Nieto, & López-Palacios (2015) study funding success, finding that the reported loan use is a significant factor explaining differences in default rates. Specifically, the authors argue that there is heterogeneity in the credit riskiness of individuals seeking specific loan purposes, indicating that credit rating can explain some variation in borrowers’ choice of loan use. Therefore, we formulate the following hypothesis:

Hypothesis 1: *Borrowers with the same credit ratings choose similar loan uses*

Supporting the familiarity bias, Ravina (2019) studies personal characteristics in P2P markets, finding that loans tend to be awarded to those sharing personal characteristics such as occupation. As Ayal et al. (2018) implies, this can be generalised to suggest that lenders’ loan grants reflect a wider network of similar borrower preferences, which can reflect the homogeneity in loan use

observed by Serrano-Cinca et al. (2015). Thus,

Hypothesis 2: Borrowers sharing the same occupation area tend to share loan uses

Further, behavioural finance suggests that individuals' financial decisions are not independent, but rather, shaped by bounded rationality and psychological preferences (Kahneman & Tversky, 1979). In the P2P context, Ayal et al. (2018) shows that these preferences can influence how borrowers choose loan types and how this can overlap with other borrowers' decisions. Using the Exponential Random Graph Model (ERGM) framework, we can identify how these behaviours manifest as structural regularities in a borrower-to-loan-use network. Therefore, we ask:

Research Question 2: *What behavioural mechanisms affect the structure of borrower-to-loan-use connections?*

The aforescribed section outlined that the familiarity bias can lead to similarities across borrower portfolios in terms of their characteristics. However, in P2P settings, this can manifest itself as familiarity *clustering*, wherein borrowers who have previously been granted certain loan types are more likely to repeat their choices, or select other categories that appear familiar to different borrowers. Herzenstein et al. (2011), studying borrower decisions on Prosper.com, identifies a 'strategic herding' behaviour, where individuals gravitate towards popular or socially-validated loans. Considering that loan popularity is directly observable through its number of bids, we assert that borrowers linked to one loan purpose are likely to connect with others sharing the same loan use. In network terms, this behaviour results in cyclical substructures where there are overlapping nodes of each of the bipartite partitions. We capture a conservative form of this cycle through the ERGM term `cycle(4)`, formulating:

Hypothesis 3: Borrowers who share one loan purpose are likely to share another

Further, Ayal & Zakay (2009), expands on the theory of Debt Account Aversion with the Theory of Perceived Diversification, explaining that borrowers' distress increases with not just by the number of debts, but also the perceived *distinctiveness* of each debt type, suggesting that in the P2P context, debt mentally differentiated by purpose would also be seen as distressing. Therefore, we assert that borrowers try to minimise loan distinctiveness by requesting loans of a single type. This can be captured by the `b1degree(1)` term, which models whether borrowers are only linked to one loan purpose. Therefore, we formulate:

Hypothesis 4: Borrowers exhibit preferences for minimal debt distinctiveness, selecting one loan use

Notwithstanding, credit choices relate to non-structural factors. Cooper, Gorbachev, & Luengo-Prado (2023) show that younger borrowers face constraints in the types of loans they can access in typical credit markets. The authors argue that this constraint shapes how these borrowers form loan portfolios, often concentrating around different use cases such as education, personal consumption, or small business loans. Considering the systemic differences in how age affects credit consumption, we assert that younger borrowers have different loan use portfolios. The ERGM term `b1cov("age")` can capture this by assessing how changes in age affect how borrowers form connections with different loan types. Therefore:

Hypothesis 5: Younger borrowers have different loan use mixes than older borrowers

Alongside age, the borrower's gender is known to be a significant factor in influencing credit use decisions. Aliano, Alnabulsi, Cestari, & Ragni (2023) study the role of gender and other factors on

borrowers’ probability of default on the Bondora.com, finding a persistent gender effect on different loan uses; women tend to default less on health, home, and business loans. Croson & Gneezy (2009) shows that these differences can be due to risk perception and self-selection effects, where women exhibit greater risk aversion, leading to different borrowing patterns. We expect that these differences in risk tolerance and perceived creditworthiness by lenders affect loan use composition in terms of gender. The term `b1nodematch("gender")` captures this effect by assessing homophily in loan use selection. Our premise is supported by a negative estimate for the term, indicating heterophily. Therefore:

Hypothesis 6: Gender differences lead to different loan use mixes

Table 1: Hypotheses related to Study 2

Hypothesis	Term	Explanation
H_3^a : Borrowers who share one loan purpose are likely to share another	<code>cycle(4)</code>	This captures the tendency of borrowers and loan uses to be cyclical, where people cluster around shared uses of loans.
H_4^a : Borrowers exhibit preferences for minimal debt distinctiveness, selecting one loan use	<code>b1degree(1)</code>	This captures the tendency of borrowers to prefer minimal distinctiveness in their loan uses, preferring to hold only one distinct type to minimise their perceived risk.
H_5^a : Younger borrowers have different loan use mixes than older borrowers	<code>b1cov("age")</code>	This captures the tendency for younger borrowers to choose different kinds of loans based on how differently they consume credit.
H_6^a : Gender differences lead to different loan use mixes	<code>b1nodematch("gender")</code>	This captures the idea that there are systemic differences in what kinds of loan uses are preferred the genders due to their perceived riskiness and creditworthiness.

Following this section, we outline the methodology, explaining our dataset and network construction. To facilitate this, descriptive statistics are analysed for both the dataset and resulting network. Subsequently, we elaborate on our choice of models, discussing how network models can be used to conduct the study. Then, modelling results are shown and explained to evaluate our hypotheses. Simultaneously, the robustness of the study is evaluated through the models’ diagnostics. Finally, we arrive at our conclusions, summarising the study, its results, and implications. Supporting items such as the source code and specific data processing steps are shown in section A.

2 Dataset

The study utilises publicly-available data from Bondora, a European P2P lending platform primarily operating in the Baltics and Spain. The dataset contains detailed information on defaulted

and non-defaulted loans granted to users between February 2009 and July 2021. Specifically, contained is a range of numeric, binary, categorical, and time-series attributes across 85,087 unique users and 179,235 individual loans. Since the company’s API is no longer accessible, we utilised a publicly available repository compiled by Manu Siddhartha (2021) on kaggle.com. The user made a series of API calls to collect the data. Following this, the dataset was processed according to Appendix A.1.

2.1 Pre-Processing & Network Formation

Initially, the dataset was filtered loans between 2014 and 2016 since the data is incomplete during other periods. Since we are studying interactions across two distinct set of nodes, a bipartite network was formed with unique users as the first partition and the purpose of loans as the second. This allows us to evaluate how borrowers interact with loan purposes; a one-mode projection onto users is not meaningful for P2P networks as it produces no discernible structure. For model efficiency and performance, we randomly sample 500 individuals from the larger sample. Ultimately, we have 500 and 9 nodes in the first and second partitions, respectively.

2.2 Descriptive Statistics & Preliminary Analysis

Table 2 summarises the numeric variables between the reduced and larger samples. First, borrowers take out relatively small loans with fairly high interest rates, possibly reflecting that the average borrower is deemed fairly risky. The average loan is quite long, at approximately 46 months out of a maximum 60. Figure 11 shows that the largest credit category is HR¹, which is the lowest possible rating. So, the typical borrower is relatively uncreditworthy. Moreover, the largest categories of loan use are “Other” and “Home Improvement”, suggesting these are largely personal. We observe minimal differences between the random sampling and the larger sample of individuals, per Table 2 and Figure 11, however, the study may be biased by the fact that the dataset contains only granted loans, rather than all bids for loans. This means our inferences about how borrowers behave is limited to successful bids. Further, the time period is relatively distant, meaning that recent structural changes to the P2P sector cannot be accounted for.

Table 2: Descriptive Statistics of Numeric Variables

DataFrame	Variable	Mean	Median	Std. Dev.	Min	Max
Initial Sample	Amount	2652.01	2125	2151.28	115	10630
Initial Sample	Interest	35.94	31	24.77	7.62	263.63
Initial Sample	Age	38.53	37	11.4	19	70
Initial Sample	LoanDuration	45.5	60	17.53	3	60
Reduced Sample	Amount	2637.64	2112.5	2112.2	170	10630
Reduced Sample	Interest	35.85	30	27.24	10.17	253.08
Reduced Sample	Age	38.73	37	11.34	21	69
Reduced Sample	LoanDuration	46.23	60	17.35	3	60

¹Bondora uses their own proprietary credit rating system where the best rating is AA and worst is HR (Bondora, 2024). The company bases these ratings on their calculated expected probability of loss on the loan. For example, someone rated AA ranges from a expected loss of 0% to 2%, whereas for HR this is 25% - >25%.

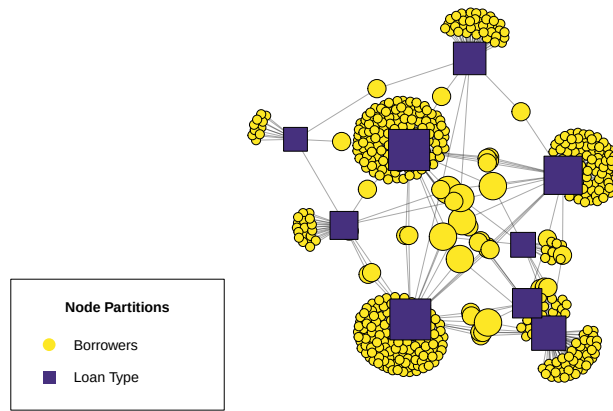
Table 3 shows a network density of 0.12, indicating that only 12% of all possible borrower–loan-type connections exist. This density suggests that borrowers typically engage with few loan types, reflecting specialization in borrowing behaviour. The centralization value of 0.47 further indicates an uneven distribution across loan types, where a few popular categories attract many borrowers while most remain peripheral. The mean distance of approximately three implies that borrowers are closely connected, typically separated by only three-to-four loan-use steps.

Centrality distributions reinforce this structure Figure 2. Betweenness is highest among loan-type nodes, though several borrowers also bridge distinct loan clusters. Closeness centrality remains low but scattered, suggesting localized clustering constrained by the bipartite structure. Degree distributions show wide variation among loan types, consistent with a hub-like structure dominated by a few popular uses. However, the second partition has relatively small degree dispersion. Finally, the homogeneous eccentricity distribution indicates that no borrowers are highly isolated, reflecting a compact and moderately cohesive network overall.

Table 3: Basic Network Descriptive Statistics

Statistic	Measure
Vertex Count	509.00
Edge Count	547.00
Density	0.12
Centralization	0.47
Mean Distance	3.66

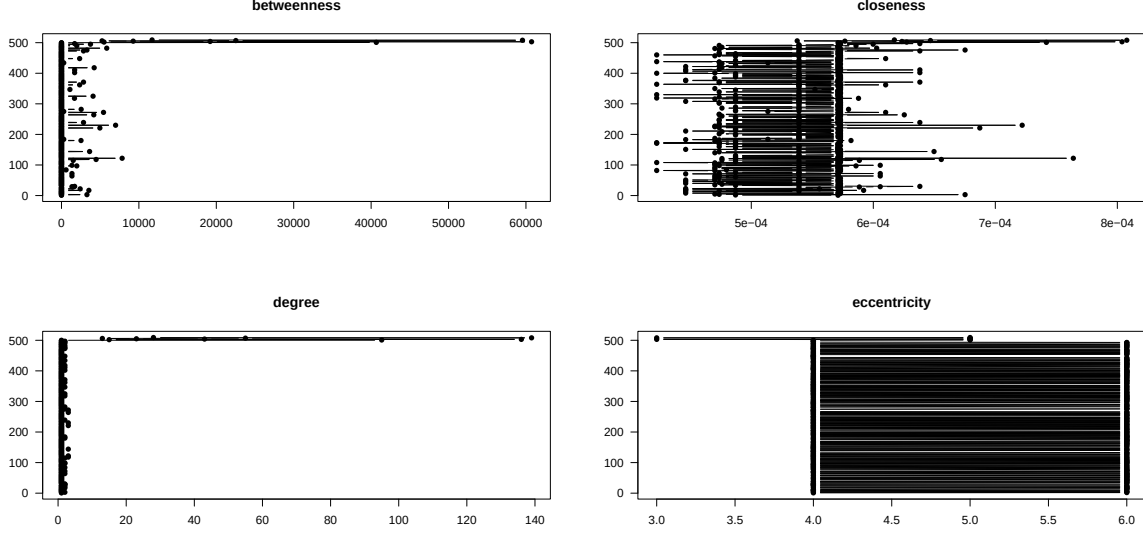
Figure 1: Network Visualisation



Note: Each colour represents a different partition of the network. The partitions' nodes are scaled to their relative degrees. Users with many different loan types are larger than those with single loan uses. Similarly, the size of second partition

nodes represents how many users belong to each.

Figure 2: Network Centrality Distribution Plots



3 Research Rationale

Our overall study analyses borrowers' behaviours, however, the studies are differentiated in their scope; the first study observes dyad-level relations among borrowers to understand how shared attributes affect borrowers' behaviours. This is inherently useful because behavioural finance aims to understand how individuals behave in *aggregate* (Ayal et al., 2018). Typically, the Quadratic Assignment Procedure is selected for dyad-level observations because basic linear regressions such as Ordinary Least Squares are fundamentally flawed and biased when using non-independent observations (Simpson, 2001).

Recognising this, studies such as Zuo (n.d.) assume that groups of borrowers have dependencies, and accordingly, use clustering methods such as K-Means or Fuzzy Clustering to understand behaviour. However, these methods are unsuitable when we require a distribution of estimates to make inferences from our results. Moreover, studying individuals as dyads rather than large clusters provides finer granularity in results. Therefore, QAP regression is the most appropriate choice for this study: it allows us to control for the non-independence among dyads, derive a statistical distribution of estimates through matrix permutation, and draw valid inferences about the relationship between borrower similarities and loan-use behaviour. This makes it both theoretically and methodologically well-suited to our research objectives and dataset.

The second study extends the analysis from a dyadic similarities to the structural formation of borrower-to-loan-use connections, which can identify the behavioural mechanisms that influence borrowers' specific loan use mixes. While QAP can account for exogenous influences, it cannot analyse higher-order network patterns such as clustering while accounting for simultaneous interactions between borrowers and loan uses. Using the Exponential Random Graph Model allows us to analyse how local behavioural tendencies aggregates to a global network structure.

Recent evidence highlights the need to focus on structural approaches in P2P lending. Liu, Baals, Osterrieder, & Hadji-Misheva (2024) studies borrowers’ centrality with the Bondora P2P dataset, modelling edges through borrower-to-borrower similarity metrics. Ultimately, they find that a borrower’s position is highly significant for their probability of default, using centrality as an exogenous variable in a logistic regression. In contrast, ERGM allows us to move away from predictive association and instead account for endogenous factors, allowing us to understand the likelihood that borrowers’ selection of loan uses are random. Consequently, we can better understand how behavioural phenomena such as familiarity and debt account aversion can translate into *systemic* patterns of credit use.

As we aim to understand how two distinct groups of nodes interact, a bipartite approach is highly suitable. Specifically, Stivala, Wang, & Lomi (2025) shows that in bipartite networks, terms such as cycle(4) can be effective in capturing clustering as in hypothesis 3, whereas it would fail in one-mode projected networks. Furthermore, the flexibility of ERGM also allows for modelling exogenous impacts such as heterophily in gender-based loan use mixes, as in hypothesis 6, and covariate age effects as in hypothesis 5.

4 Results

4.1 Study 1 - QAP Linear Regression

Figure 3

Figure 3: QAP Coefficient Plots

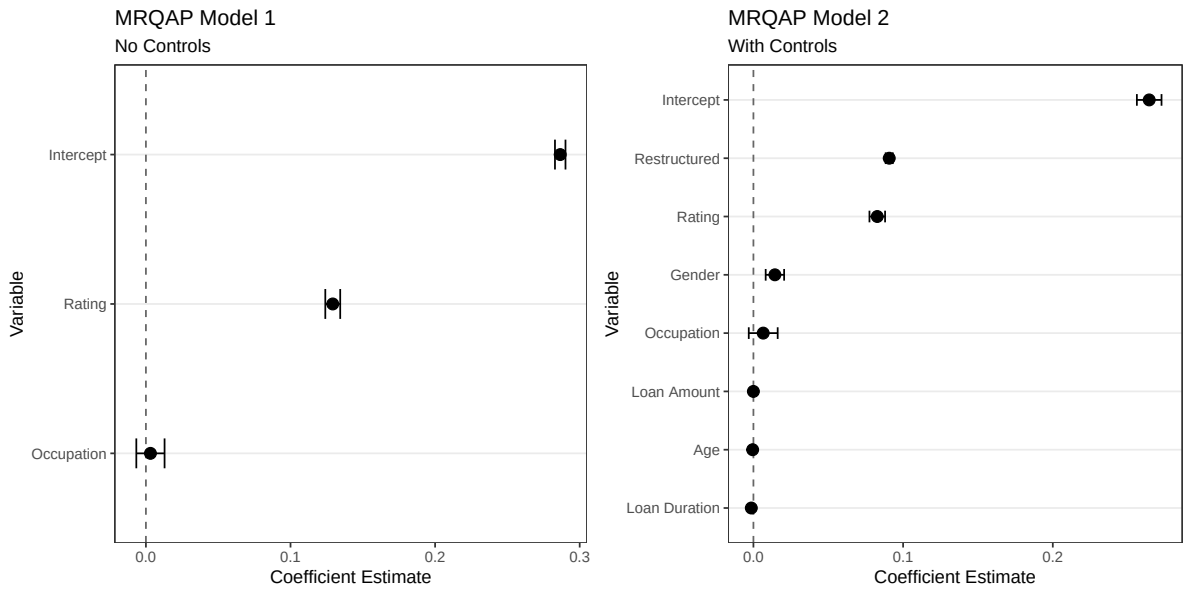


Table 5

Table 4

Table 5: QAP Modelling Results

	M1 No Controls	M2 Controls
Intercept	0.287*** (0.002)	0.264*** (0.004)
Rating	0.129*** (0.003)	0.083*** (0.003)
Occupation	0.003 (0.005)	0.007 (0.005)
Loan Amount		−0.000** (0.000)
Age		−0.001 (0.000)
Gender		0.014 (0.003)
Loan Duration		−0.001** (0.000)
Restructured		0.091*** (0.001)
R-squared	0.010	0.035

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 6

Table 7: ERGM Modelling Results

	Base ERGM	b1degree(1)	cycle(4)	Age	Gender
edges	-1.978*** (0.046)	-1.354*** (0.123)	-0.940*** (0.279)	-0.862** (0.291)	-0.538 (0.462)
b1degree1		3.309*** (0.224)	3.267*** (0.250)	3.260*** (0.244)	3.271*** (0.247)
cycle4			-0.141 (0.087)	-0.141 (0.086)	-0.147 (0.086)
b1factor.b1_gender.male				-0.181 (0.207)	-0.168 (0.203)
b1cov.b1_age					-0.008 (0.009)
AIC	3325.828	2674.435	2676.439	2672.013	2674.852
BIC	3332.238	2687.255	2695.668	2697.652	2706.901
Log Likelihood	-1661.914	-1335.218	-1335.219	-1332.006	-1332.426

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

4.2 Study 2 - ERGM

Figure 4

Figure 4: ERGM Coefficient Plots

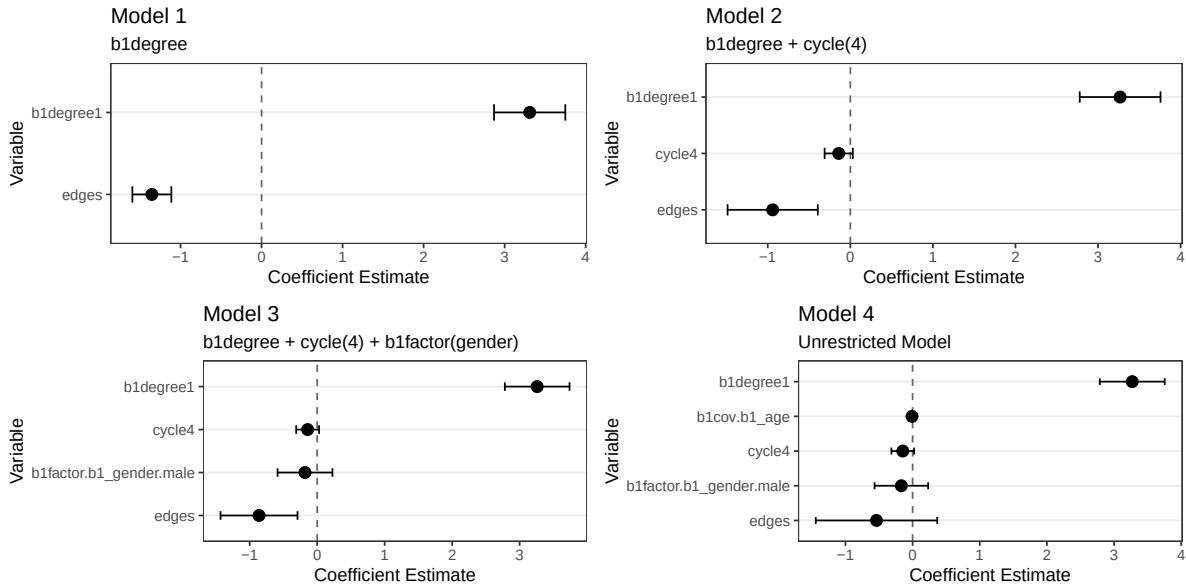


Table 7

A Supplements

A.1 Data Preprocessing Steps

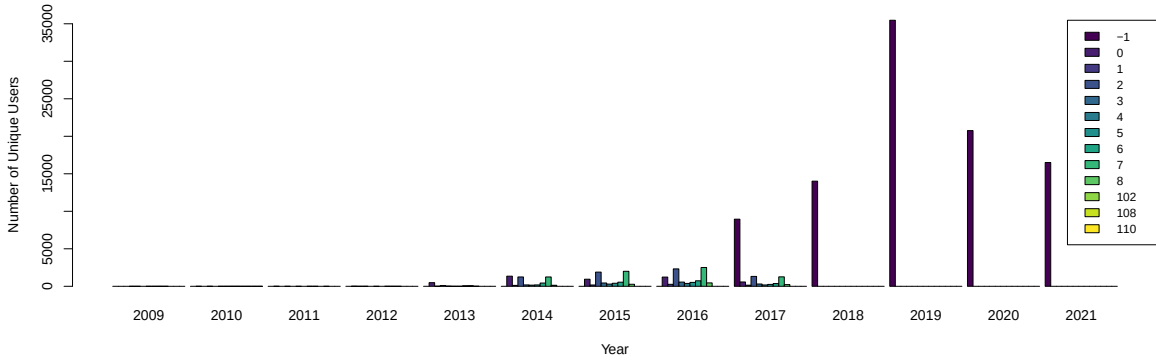
This section outlines the specific steps taken to process the data into the network object that was used in later analysis, outlining the rationale, where needed.

- Following the data import, only the following attributes were kept to make our processing steps more focused

```
keep_cols <- c("LoanId", "UserName", "Age", "Gender",  
              "Country", "Amount", "Interest", "LoanDuration",  
              "UseOfLoan", "Rating", "Restructured", "MonthlyPayment",  
              "OccupationArea", "BiddingStartedOn")
```

- We removed rows with any NA values for the selected attributes to ensure that we have a complete dataset
- Observing Figure 5, we see that between 2017-2021, there exists many -1 values for UseofLoan. This is because Bondora no longer provided this detail in later datasets. However, between 2014 and 2016 this data is available. Because of this, we restrict the time period to have the least noisy data. BiddingStartedOn was chosen to record the time, because the auction start date can be a good indication of activity around the loan.

Figure 5: Number of Unique Users per Loan Type per Year



- Then, we sampled 500 individuals out of the filtered dataset to ensure that our models run sufficiently well and converge. By choosing 500 individuals, we preserve importance structures within the network while adding sufficient statistical power.
- When adding labels to each loan use and occupation area, we keep any possible missing values because these do not indicate lower quality data. Instead, they can carry information about lenders distribute loans according to the quality of information presented by borrowers (Yao et al., 2019). Additionally, keeping missing labels is important to ensure that the edges are not superficially limited and that nodes can be isolates if they are indeed that way in reality.
- To create the network object for ERGM models:

- We first create an incidence matrix of size $n \times m$, borrower usernames by the loan use, where the values are the number of loans the user obtained for each loan use category. These values are then turned into binary because our modelling does not support the use of weighted edges.
- A bipartite network is created from this incidence matrix, where the first partition are unique users, and the second partition is the loan use associated among all loans of the user. Users can share more than more loan use type.
- We add age and gender attributes to each node of the first partition
- To create the relevant networks/matrices for QAP Regressions:
 - We first re-create the loan use incidence matrix, however, we keep the counts as the linear regression supports this.
 - * We then convert the loan use incidence matrix into an adjacency matrix of shape $n \times n$ through the transformation $\mathbf{X} \cdot \mathbf{X}^T$, which results in a matrix of shared loan uses weighted by the frequency of each user’s loan count within each loan use category.
 - * We then set the diagonal values of $\mathbf{X}$ to be zero to ensure that there are no loops within the network.
 - In accordance with this procedure, we create adjacency matrices for Credit Rating and Occupation Area which will be used as the main predictors in the QAP models.
 - * Credit Rating is a weighted adjacency matrix based on the loan count belonging to each user for each credit rating category
 - * Occupation Area is a binary adjacency matrix based on which occupation each user has reported to have held in the past when applying for each of their loans.
 - Additionally, we create a set of control variables to improve the validity and performance of the linear regression using Loan Amounts, Age, Gender, Loan Duration, and whether the loans have been restructured.
 - * Loan Amounts is a weighted adjacency matrix based on five bins across the range of possible loan amounts
 - * Age is an adjacency matrix based on differences in users’ ages
 - * Gender is a binary adjacency matrix based on users’ shared gender.
 - * Loan Duration is an adjacency matrix based on the differences in users’ loans average durations
 - * Restructured is a binary adjacency matrix based on whether the users share the fact that they have defaulted on any loans in the past. The users can either have shared both defaults and non-defaults, or both.

A.2 Distributions of Variables across Samples

Figure 6: Distribution of Loan Amount across Samples

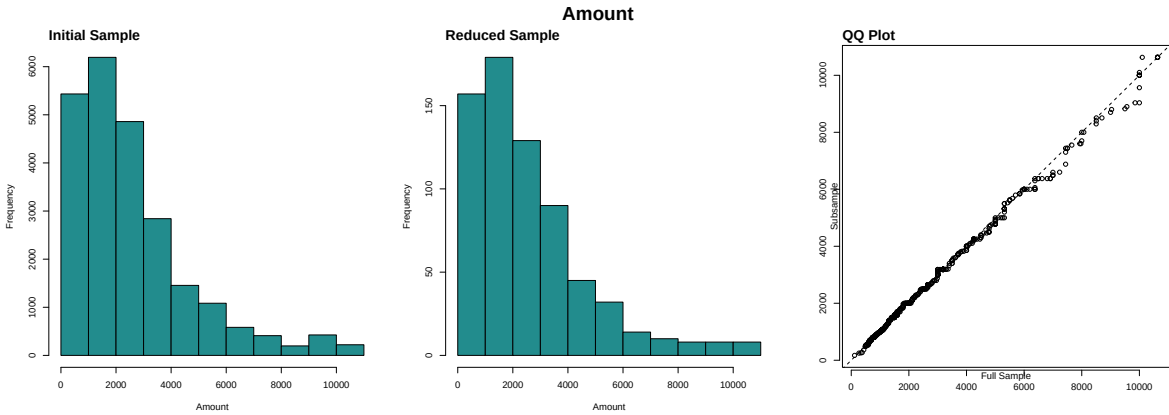


Figure 7: Distribution of Interest Rates across Samples

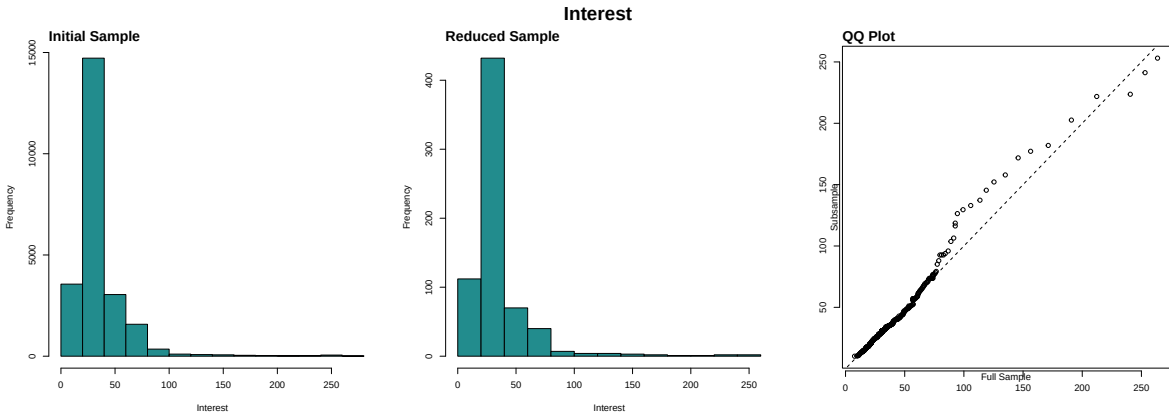


Figure 8: Distribution of Age across Samples

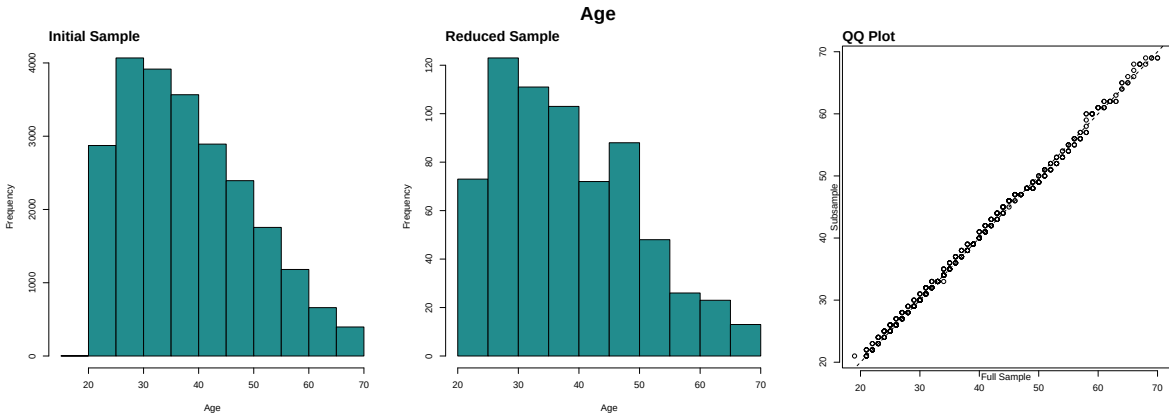


Figure 9: Distribution of Loan Duration across Samples

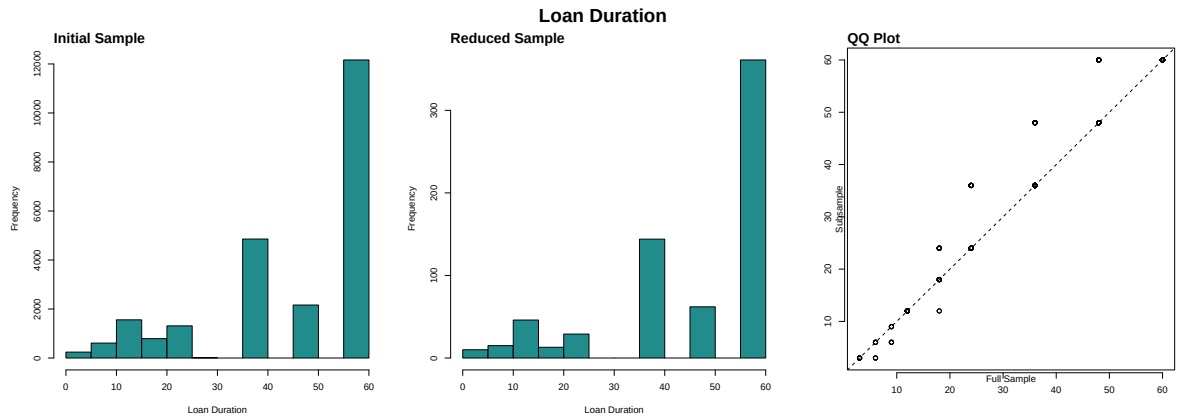


Figure 10: Distribution of Loan Purpose across Samples

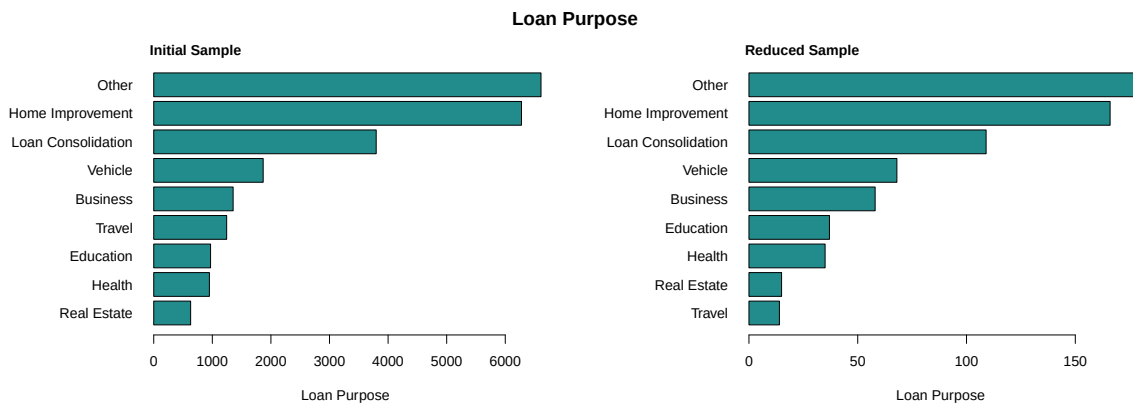


Figure 11: Distribution of Credit Ratings across Samples

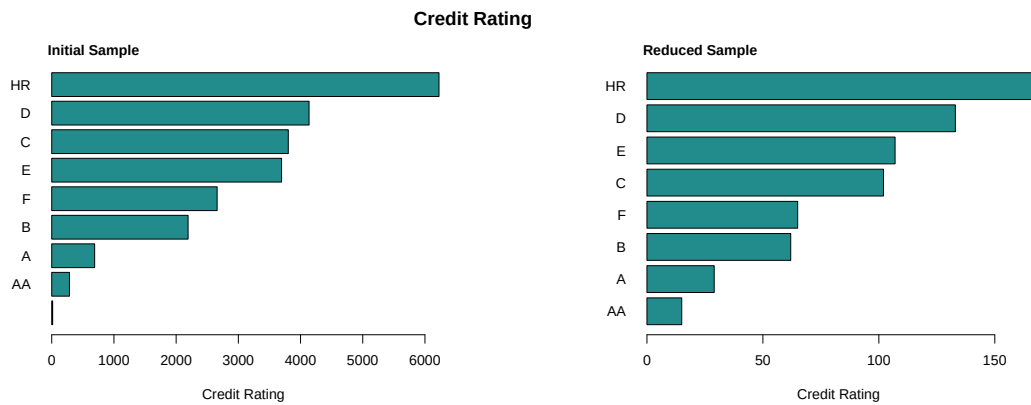
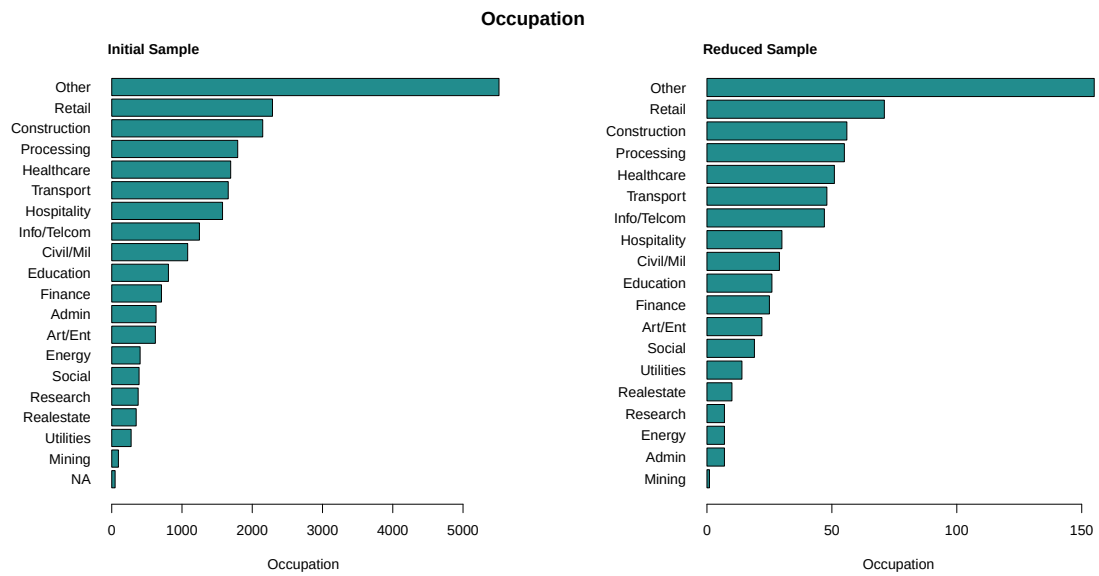


Figure 12: Distribution of Occupation Types across Samples



A.3 ERGM Results - Supplementary Tables

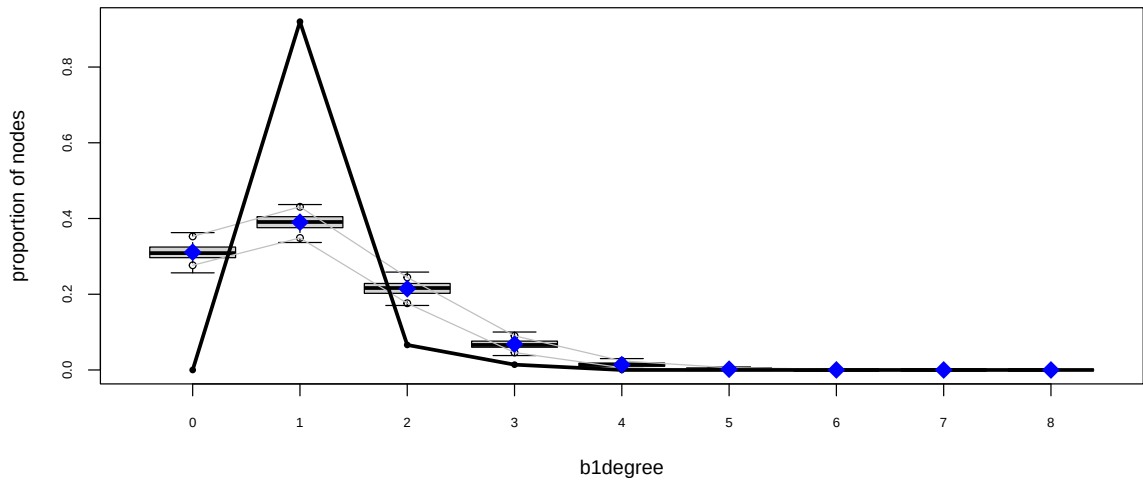
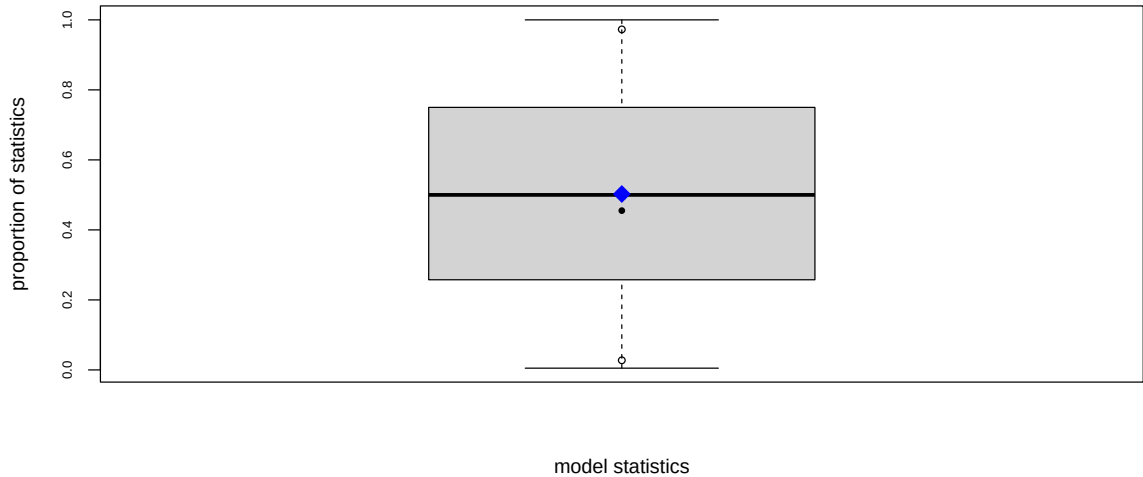
Table 8: ERGM Results in Probability Scale

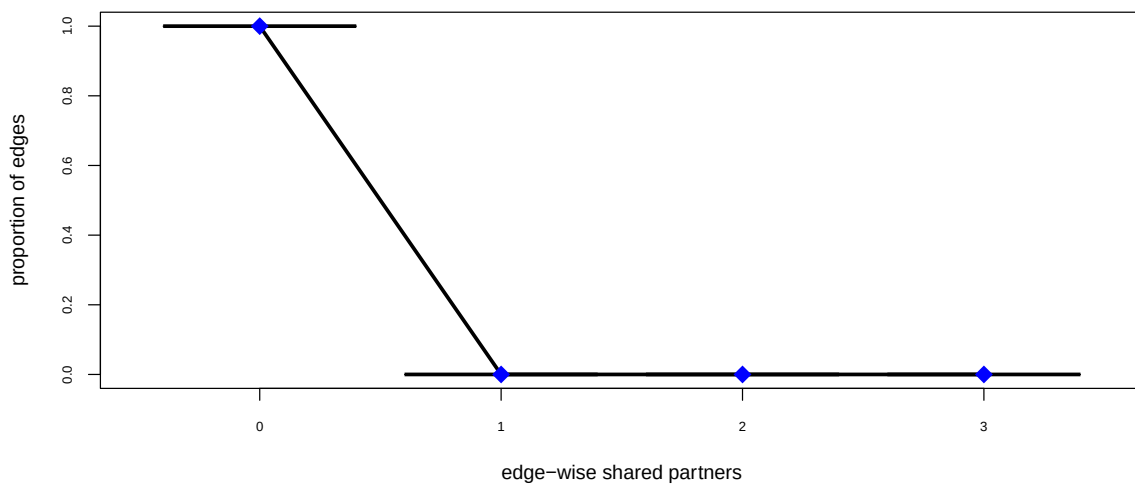
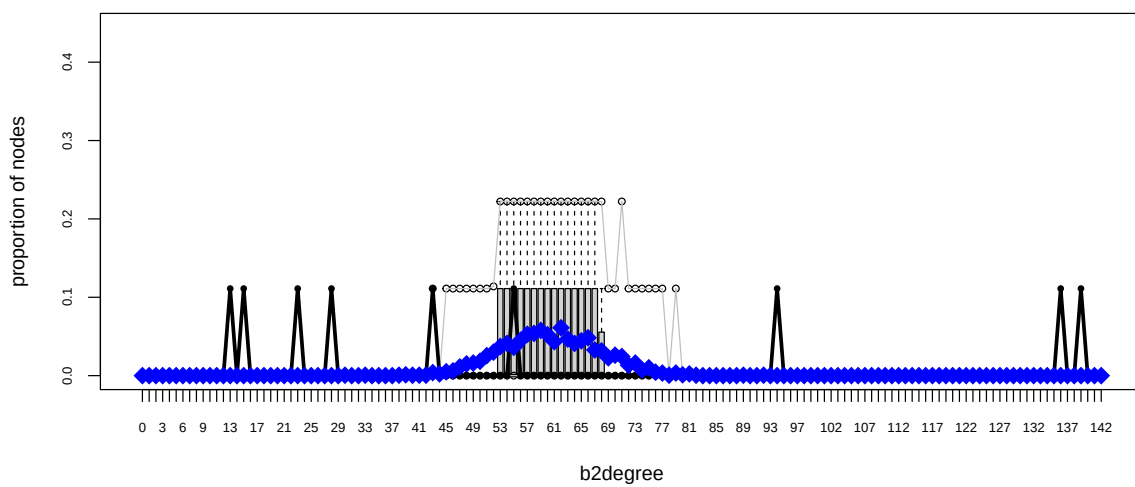
Terms	Base	b1degree.1.	cycle.4.	Age	Gender
edges	0.122	0.205	0.281	0.297	0.369
b1degree1	-	0.965	0.963	0.963	0.963
cycle4	-	-	0.465	0.465	0.463
b1factor.b1_gender.male	-	-	-	0.455	0.458
b1cov.b1_age	-	-	-	-	0.498

A.4 ERGM Diagnostics Plots

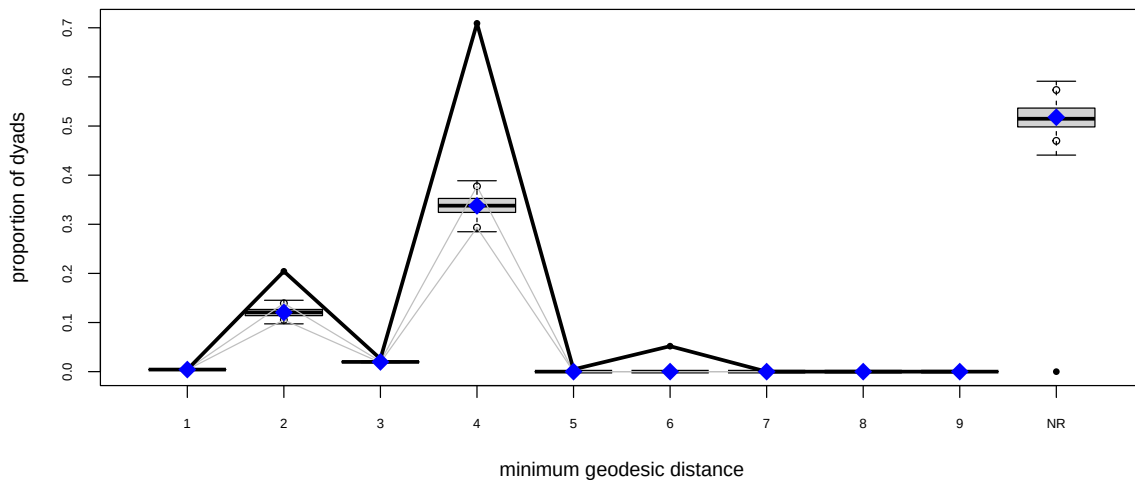
A.4.1 Base Model

```
Registered S3 methods overwritten by 'snafun':
  method      from
plot.network  network
print.network network
```





Goodness-of-fit diagnostics



Goodness-of-fit for model statistics

obs	min	mean	max	MC p-value
546.00	487.00	543.86	602.00	0.91

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max	MC p-value
1	546	487	543.860	602	0.91
2	26301	12531	15520.635	18711	0.00
3	3351	2348	2549.430	2749	0.00
4	91309	36680	43470.795	50047	0.00
5	594	0	0.460	48	0.00
6	6677	0	5.255	567	0.00
Inf	0	56768	66687.565	76128	0.00

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max	MC p-value
0	0	128	155.250	183	0.00
1	459	168	194.705	230	0.00
2	33	73	107.065	129	0.00
3	7	19	34.040	50	0.00
4	0	2	6.865	15	0.00
5	0	0	1.005	4	0.75
6	0	0	0.070	1	1.00

Goodness-of-fit for bipartition 2 degree

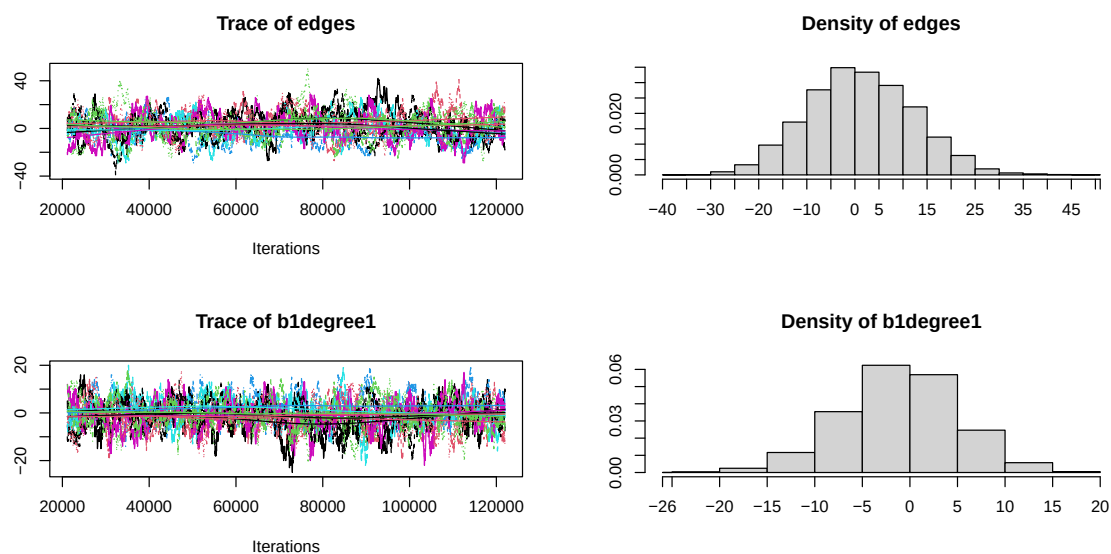
	obs	min	mean	max	MC	p-value
13	1	0	0.000	0		0.00
15	1	0	0.000	0		0.00
23	1	0	0.000	0		0.00
28	1	0	0.000	0		0.00
30	0	0	0.005	1		1.00
33	0	0	0.005	1		1.00
38	0	0	0.005	1		1.00
39	0	0	0.010	1		1.00
40	0	0	0.005	1		1.00
41	0	0	0.005	1		1.00
42	0	0	0.005	1		1.00
43	1	0	0.035	1		0.07
44	0	0	0.020	1		1.00
45	0	0	0.045	1		1.00
46	0	0	0.055	1		1.00
47	0	0	0.100	1		1.00
48	0	0	0.135	2		1.00
49	0	0	0.145	2		1.00
50	0	0	0.165	2		1.00
51	0	0	0.230	2		1.00
52	0	0	0.270	2		1.00
53	0	0	0.335	3		1.00
54	0	0	0.370	3		1.00
55	1	0	0.325	2		0.56
56	0	0	0.405	3		1.00
57	0	0	0.475	3		1.00
58	0	0	0.485	2		1.00
59	0	0	0.520	3		1.00
60	0	0	0.470	3		1.00
61	0	0	0.390	2		1.00
62	0	0	0.550	3		1.00
63	0	0	0.420	3		1.00
64	0	0	0.370	3		1.00
65	0	0	0.400	3		1.00
66	0	0	0.435	4		1.00
67	0	0	0.295	3		1.00
68	0	0	0.285	3		1.00
69	0	0	0.205	2		1.00
70	0	0	0.235	2		1.00
71	0	0	0.220	2		1.00
72	0	0	0.120	2		1.00
73	0	0	0.145	3		1.00
74	0	0	0.065	1		1.00
75	0	0	0.090	1		1.00

76	0	0	0.040	1	1.00
77	0	0	0.030	1	1.00
78	0	0	0.005	1	1.00
79	0	0	0.030	1	1.00
80	0	0	0.010	1	1.00
81	0	0	0.020	1	1.00
82	0	0	0.005	1	1.00
89	0	0	0.005	1	1.00
92	0	0	0.005	1	1.00
94	1	0	0.000	0	0.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max MC	p-value
546.00	487.00	543.86	602.00	0.91

A.4.2 Model 1



Sample statistics summary:

Iterations = 21136:121992
 Thinning interval = 28
 Number of chains = 9
 Sample size per chain = 3603

1. Empirical mean and standard deviation for each variable,
 plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
edges	1.727	11.066	0.06145	0.6592
bldegree1	-0.554	6.039	0.03353	0.2513

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
edges	-19	-6	1	9	24
bldegree1	-13	-4	0	3	11

Are sample statistics significantly different from observed?

	edges	bldegree1	(Omni)
diff.	1.726955932	-0.55404447	NA
test stat.	2.577412722	-2.16661869	5.80269221
P-val.	0.009954302	0.03026394	0.05651482

Sample statistics cross-correlations:

	edges	bldegree1
edges	1.0000000	-0.6736048
bldegree1	-0.6736048	1.0000000

Sample statistics auto-correlation:

Chain 1

	edges	bldegree1
Lag 0	1.0000000	1.0000000
Lag 28	0.9826388	0.9701176
Lag 56	0.9664557	0.9439196
Lag 84	0.9498574	0.9180981
Lag 112	0.9335225	0.8938654
Lag 140	0.9173206	0.8721985

Chain 2

	edges	bldegree1
Lag 0	1.0000000	1.0000000
Lag 28	0.9829174	0.9611738
Lag 56	0.9663965	0.9239480
Lag 84	0.9502249	0.8879717
Lag 112	0.9333931	0.8535116
Lag 140	0.9178077	0.8205518

Chain 3

	edges	bldegree1
Lag 0	1.0000000	1.0000000
Lag 28	0.9883845	0.9692833
Lag 56	0.9772397	0.9397716
Lag 84	0.9662077	0.9126215

Lag 112 0.9556828 0.8871831
Lag 140 0.9453678 0.8626915
Chain 4

edges b1degree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9825875 0.9674543
Lag 56 0.9666999 0.9364074
Lag 84 0.9515609 0.9049248
Lag 112 0.9362933 0.8735428
Lag 140 0.9216203 0.8433670

Chain 5
edges b1degree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9817720 0.9637732
Lag 56 0.9642654 0.9307404
Lag 84 0.9471522 0.8990125
Lag 112 0.9306044 0.8686574
Lag 140 0.9141629 0.8394052

Chain 6
edges b1degree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9840958 0.9657033
Lag 56 0.9685792 0.9335799
Lag 84 0.9533266 0.9036673
Lag 112 0.9378353 0.8764645
Lag 140 0.9228011 0.8498216

Chain 7
edges b1degree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9852593 0.9626353
Lag 56 0.9707885 0.9284125
Lag 84 0.9574817 0.8962185
Lag 112 0.9445881 0.8654185
Lag 140 0.9318521 0.8363137

Chain 8
edges b1degree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9806314 0.9686263
Lag 56 0.9615573 0.9383344
Lag 84 0.9432512 0.9091995
Lag 112 0.9247775 0.8805019
Lag 140 0.9061377 0.8532768

Chain 9
edges b1degree1
Lag 0 1.0000000 1.0000000
Lag 28 0.9795136 0.9606722

Lag 56 0.9603946 0.9238283
Lag 84 0.9417626 0.8881744
Lag 112 0.9231795 0.8544565
Lag 140 0.9036287 0.8209240

Sample statistics burn-in diagnostic (Geweke):

Chain 1

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges b1degree1
-0.8251148 1.1021806

Individual P-values (lower = worse):

edges b1degree1
0.4093064 0.2703832
Joint P-value (lower = worse): 0.5497544
Chain 2

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges b1degree1
-0.6130584 1.2345011

Individual P-values (lower = worse):

edges b1degree1
0.5398377 0.2170163
Joint P-value (lower = worse): 0.3088801
Chain 3

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges b1degree1
-0.5254825 1.8245035

Individual P-values (lower = worse):

edges b1degree1
0.59924780 0.06807598
Joint P-value (lower = worse): 0.3048263
Chain 4

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```
edges b1degree1
2.1763745 -0.6342556
```

Individual P-values (lower = worse):

```
edges b1degree1
0.02952727 0.52591401
Joint P-value (lower = worse): 0.1381523
Chain 5
```

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

```
edges b1degree1
0.05826116 -0.14359472
```

Individual P-values (lower = worse):

```
edges b1degree1
0.9535406 0.8858205
Joint P-value (lower = worse): 0.9955405
Chain 6
```

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

```
edges b1degree1
-2.2393205 0.4044089
```

Individual P-values (lower = worse):

```
edges b1degree1
0.02513507 0.68591207
Joint P-value (lower = worse): 0.07543831
Chain 7
```

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

```
edges b1degree1
0.6404571 -0.4060245
```

Individual P-values (lower = worse):

```
edges b1degree1
0.5218755 0.6847246
Joint P-value (lower = worse): 0.9039975
Chain 8
```

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges b1degree1
-0.1195638 -0.2165394

Individual P-values (lower = worse):
edges b1degree1
0.9048287 0.8285673
Joint P-value (lower = worse): 0.7900773
Chain 9

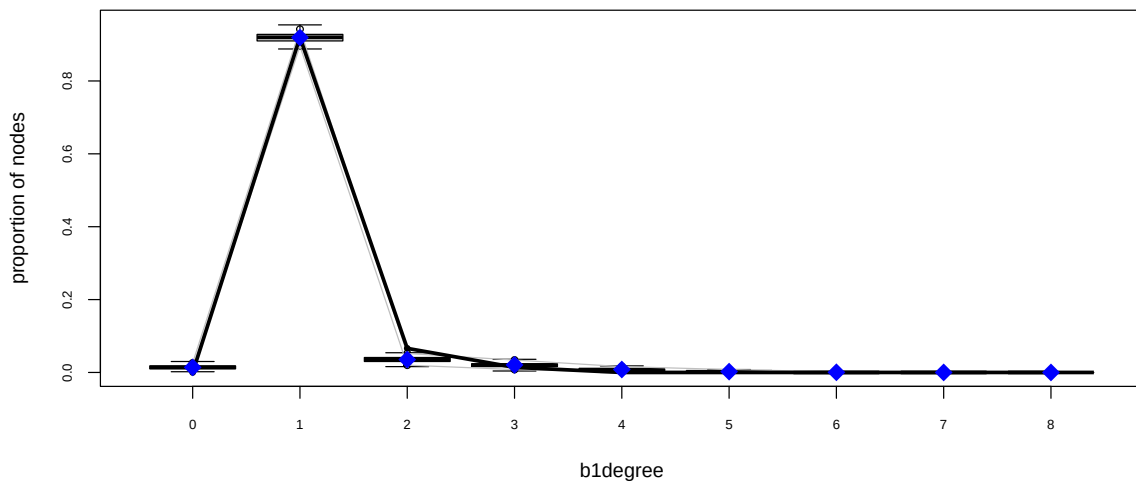
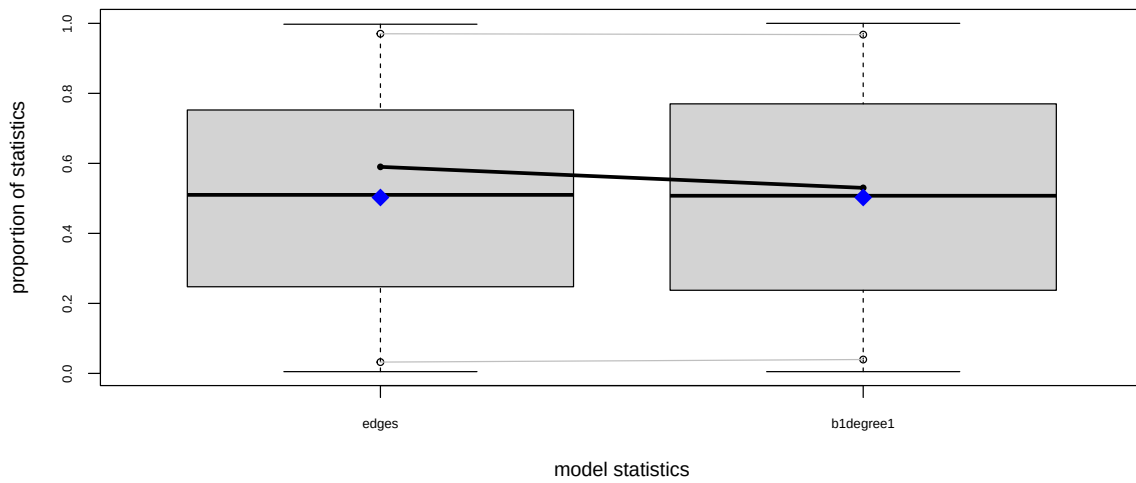
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

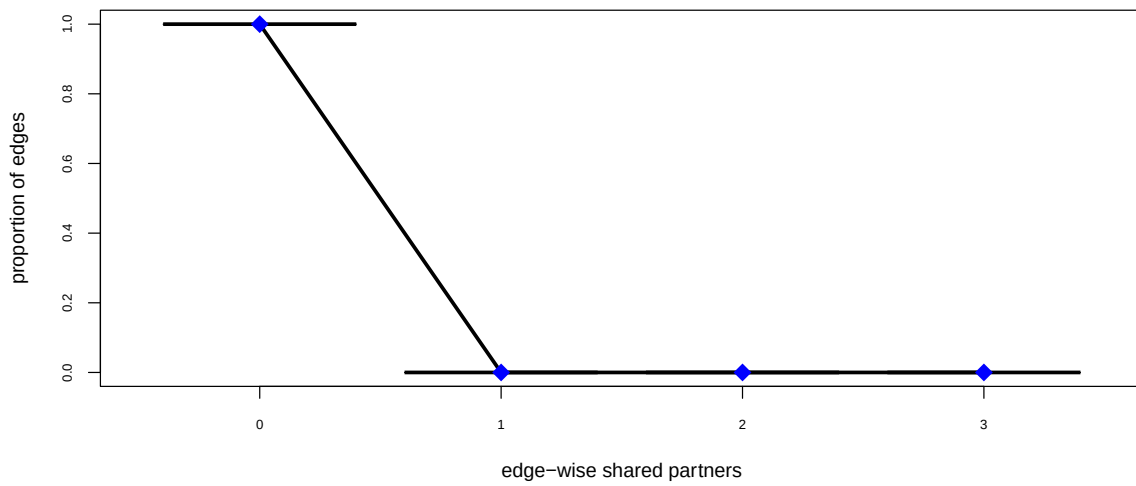
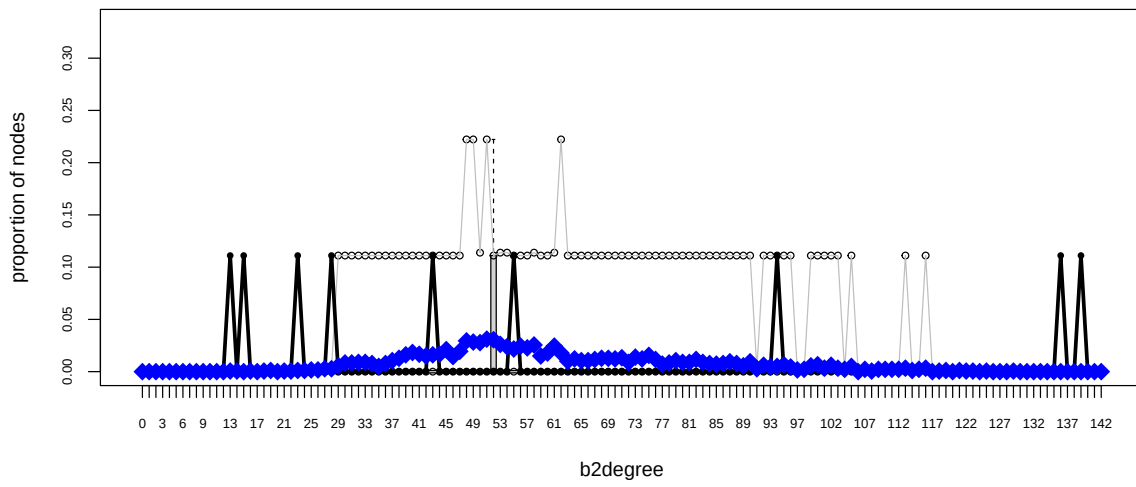
edges b1degree1
1.1789189 0.2317552

Individual P-values (lower = worse):
edges b1degree1
0.2384305 0.8167281
Joint P-value (lower = worse): 0.4940679

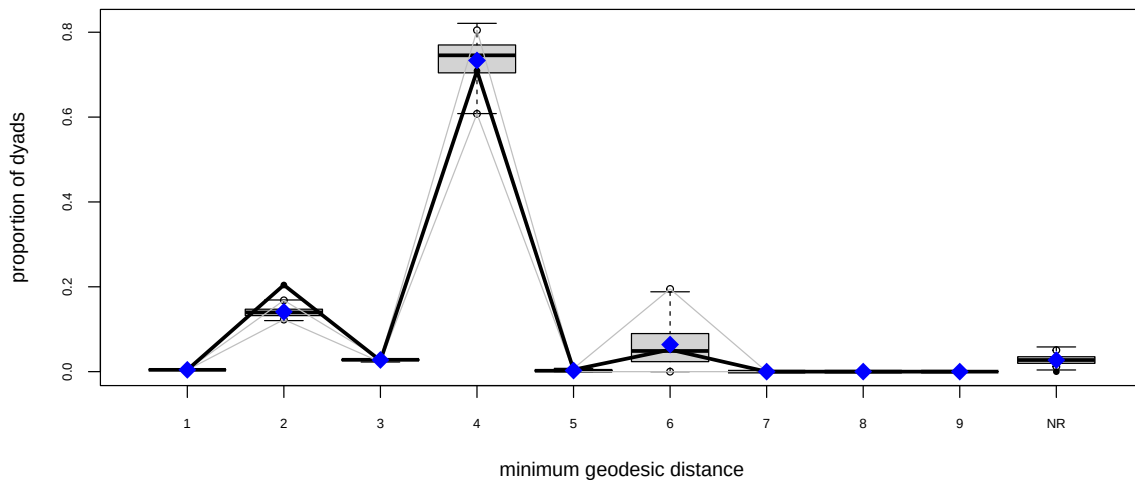
Note: To save space, only one in every 7 iterations of the MCMC sample used for estimation was stored for diagnostics. Sample size per chain was originally around 25221 with thinning interval 4.

Note: MCMC diagnostics shown here are from the last round of simulation, prior to computation of final parameter estimates. Because the final estimates are refinements of those used for this simulation run, these diagnostics may understate model performance. To directly assess the performance of the final model on in-model statistics, please use the GOF command: `gof(ergmFitObject, GOF=~model)`.





Goodness-of-fit diagnostics



Goodness-of-fit for model statistics

	obs	min	mean	max	MC	p-value
edges	546	525	548.185	574		0.86
b1degree1	459	437	458.670	477		1.00

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max	MC	p-value
1	546	525	548.185	574		0.86
2	26301	15493	18153.770	23058		0.00
3	3351	2676	3547.225	3909		0.43
4	91309	74402	94453.820	105721		0.54
5	594	0	332.150	1251		0.32
6	6677	0	8222.595	31969		0.90
7	0	0	0.800	160		1.00
8	0	0	15.000	3000		1.00
Inf	0	507	3504.455	7992		0.00

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max	MC	p-value
0	0	1	6.960	16		0.00
1	459	437	458.670	477		1.00
2	33	8	17.630	31		0.00
3	7	2	10.340	24		0.39
4	0	0	4.040	10		0.05
5	0	0	1.120	4		0.68

6	0	0	0.210	2	1.00
7	0	0	0.025	2	1.00
8	0	0	0.005	1	1.00

Goodness-of-fit for bipartition 2 degree

	obs	min	mean	max	MC	p-value
13	1	0	0.005	1		0.01
14	0	0	0.005	1		1.00
15	1	0	0.000	0		0.00
16	0	0	0.005	1		1.00
18	0	0	0.005	1		1.00
19	0	0	0.010	1		1.00
21	0	0	0.005	1		1.00
22	0	0	0.005	1		1.00
23	1	0	0.010	1		0.02
24	0	0	0.010	1		1.00
25	0	0	0.015	1		1.00
26	0	0	0.015	1		1.00
27	0	0	0.025	2		1.00
28	1	0	0.025	1		0.05
29	0	0	0.050	2		1.00
30	0	0	0.075	1		1.00
31	0	0	0.075	1		1.00
32	0	0	0.080	2		1.00
33	0	0	0.080	1		1.00
34	0	0	0.070	1		1.00
35	0	0	0.045	1		1.00
36	0	0	0.070	1		1.00
37	0	0	0.095	2		1.00
38	0	0	0.115	1		1.00
39	0	0	0.145	2		1.00
40	0	0	0.165	2		1.00
41	0	0	0.150	2		1.00
42	0	0	0.135	1		1.00
43	1	0	0.145	2		0.27
44	0	0	0.150	1		1.00
45	0	0	0.190	2		1.00
46	0	0	0.130	2		1.00
47	0	0	0.175	2		1.00
48	0	0	0.265	3		1.00
49	0	0	0.255	2		1.00
50	0	0	0.250	2		1.00
51	0	0	0.280	3		1.00
52	0	0	0.275	2		1.00
53	0	0	0.235	2		1.00

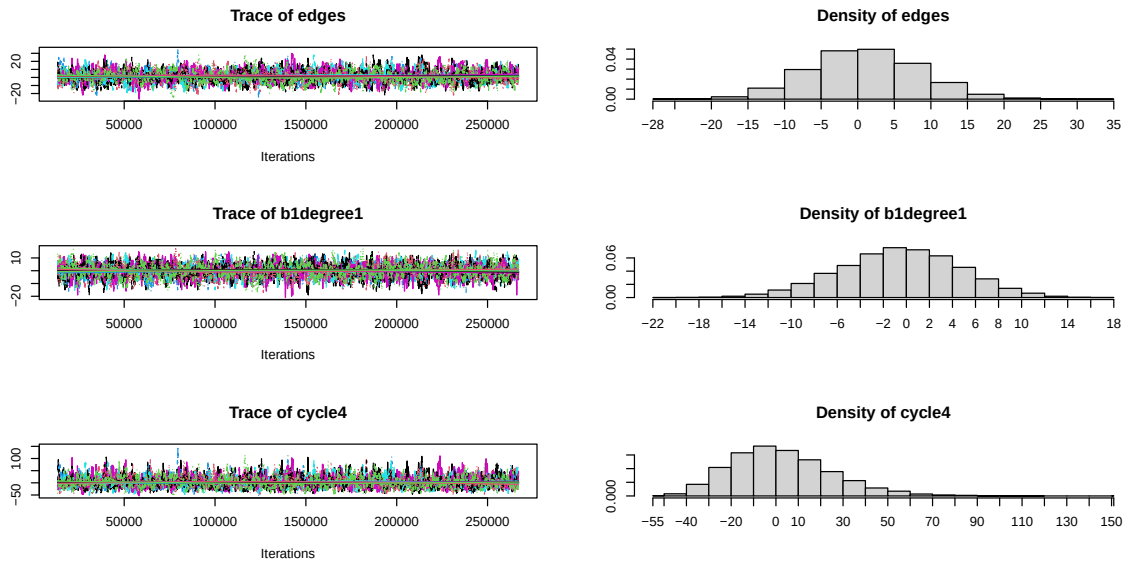
54	0	0	0.215	2	1.00
55	1	0	0.195	2	0.38
56	0	0	0.220	2	1.00
57	0	0	0.205	2	1.00
58	0	0	0.230	3	1.00
59	0	0	0.135	2	1.00
60	0	0	0.160	2	1.00
61	0	0	0.220	3	1.00
62	0	0	0.160	2	1.00
63	0	0	0.090	2	1.00
64	0	0	0.110	2	1.00
65	0	0	0.095	2	1.00
66	0	0	0.095	2	1.00
67	0	0	0.105	2	1.00
68	0	0	0.115	1	1.00
69	0	0	0.115	2	1.00
70	0	0	0.110	3	1.00
71	0	0	0.120	2	1.00
72	0	0	0.080	1	1.00
73	0	0	0.125	1	1.00
74	0	0	0.110	2	1.00
75	0	0	0.140	2	1.00
76	0	0	0.105	1	1.00
77	0	0	0.065	1	1.00
78	0	0	0.075	1	1.00
79	0	0	0.095	2	1.00
80	0	0	0.080	2	1.00
81	0	0	0.080	1	1.00
82	0	0	0.105	2	1.00
83	0	0	0.080	1	1.00
84	0	0	0.070	1	1.00
85	0	0	0.065	1	1.00
86	0	0	0.070	1	1.00
87	0	0	0.085	2	1.00
88	0	0	0.070	1	1.00
89	0	0	0.050	2	1.00
90	0	0	0.085	1	1.00
91	0	0	0.025	1	1.00
92	0	0	0.055	2	1.00
93	0	0	0.040	2	1.00
94	1	0	0.045	1	0.09
95	0	0	0.060	1	1.00
96	0	0	0.040	1	1.00
97	0	0	0.015	1	1.00
98	0	0	0.020	1	1.00
99	0	0	0.050	1	1.00

100	0	0	0.060	2	1.00
101	0	0	0.030	1	1.00
102	0	0	0.055	1	1.00
103	0	0	0.030	1	1.00
104	0	0	0.020	2	1.00
105	0	0	0.045	1	1.00
107	0	0	0.020	1	1.00
108	0	0	0.005	1	1.00
109	0	0	0.020	2	1.00
110	0	0	0.020	1	1.00
111	0	0	0.020	1	1.00
112	0	0	0.020	1	1.00
113	0	0	0.030	1	1.00
114	0	0	0.010	1	1.00
115	0	0	0.020	1	1.00
116	0	0	0.030	1	1.00
118	0	0	0.005	1	1.00
119	0	0	0.010	1	1.00
121	0	0	0.010	1	1.00
122	0	0	0.005	1	1.00
123	0	0	0.005	1	1.00
125	0	0	0.005	1	1.00
129	0	0	0.005	1	1.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max MC	p-value
546.000	525.000	548.185	574.000	0.860

A.4.3 Model 2



Sample statistics summary:

Iterations = 13376:267072
 Thinning interval = 64
 Number of chains = 9
 Sample size per chain = 3965

1. Empirical mean and standard deviation for each variable,
 plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
edges	1.44649	7.475	0.03957	0.1970
b1degree1	-0.05955	5.224	0.02766	0.1202
cycle4	3.02365	22.559	0.11942	0.5202

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
edges	-13	-4	1	7	16
b1degree1	-10	-4	0	4	10
cycle4	-34	-13	1	17	53

Are sample statistics significantly different from observed?

	edges	b1degree1	cycle4	(Omni)
diff.	1.446490e+00	-0.05954883	3.023651e+00	NA
test stat.	7.380226e+00	-0.49639643	5.879148e+00	8.382182e+01

P-val. 1.580205e-13 0.61961475 4.123823e-09 1.072914e-17

Sample statistics cross-correlations:

	edges	b1degree1	cycle4
edges	1.0000000	-0.644952	0.8593833
b1degree1	-0.6449520	1.000000	-0.5739620
cycle4	0.8593833	-0.573962	1.0000000

Sample statistics auto-correlation:

Chain 1

	edges	b1degree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9207720	0.8939305	0.8908135
Lag 128	0.8477613	0.7974903	0.7902465
Lag 192	0.7824516	0.7176731	0.7027372
Lag 256	0.7199936	0.6447281	0.6311525
Lag 320	0.6593805	0.5796761	0.5623123

Chain 2

	edges	b1degree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9127709	0.8916141	0.8699662
Lag 128	0.8323720	0.7990969	0.7576879
Lag 192	0.7585718	0.7152578	0.6632909
Lag 256	0.6928837	0.6408781	0.5816583
Lag 320	0.6307202	0.5757939	0.5101108

Chain 3

	edges	b1degree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9247032	0.9053829	0.8947725
Lag 128	0.8626373	0.8250421	0.8072021
Lag 192	0.8036997	0.7535264	0.7285900
Lag 256	0.7514687	0.6915726	0.6666843
Lag 320	0.7033289	0.6395590	0.6111047

Chain 4

	edges	b1degree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9194227	0.8882619	0.8836353
Lag 128	0.8460630	0.7872335	0.7865274
Lag 192	0.7760913	0.6949317	0.7035127
Lag 256	0.7112373	0.6160141	0.6286306
Lag 320	0.6561840	0.5565084	0.5674450

Chain 5

	edges	b1degree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9216747	0.8899834	0.8896689
Lag 128	0.8516064	0.7949439	0.7975814

Lag 192 0.7891529 0.7120355 0.7220413
 Lag 256 0.7350742 0.6384775 0.6582887
 Lag 320 0.6902214 0.5758803 0.6053976

Chain 6

	edges	b1degree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9286170	0.9040554	0.8938356
Lag 128	0.8635580	0.8200027	0.8061285
Lag 192	0.8054181	0.7433293	0.7345318
Lag 256	0.7518688	0.6798884	0.6733401
Lag 320	0.7022343	0.6197807	0.6207549

Chain 7

	edges	b1degree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9146513	0.8928942	0.8829731
Lag 128	0.8378383	0.7951906	0.7895102
Lag 192	0.7709455	0.7122812	0.7088251
Lag 256	0.7106735	0.6369230	0.6390087
Lag 320	0.6506091	0.5680243	0.5719387

Chain 8

	edges	b1degree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9169727	0.8948412	0.8835476
Lag 128	0.8384772	0.7985448	0.7808854
Lag 192	0.7696477	0.7101403	0.6929739
Lag 256	0.7056378	0.6336532	0.6158042
Lag 320	0.6465219	0.5651076	0.5422954

Chain 9

	edges	b1degree1	cycle4
Lag 0	1.0000000	1.0000000	1.0000000
Lag 64	0.9153677	0.8944287	0.8843268
Lag 128	0.8355754	0.7943154	0.7820059
Lag 192	0.7669759	0.7137630	0.6998537
Lag 256	0.7064500	0.6496772	0.6291565
Lag 320	0.6520078	0.5914224	0.5688262

Sample statistics burn-in diagnostic (Geweke):

Chain 1

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

edges	b1degree1	cycle4
-0.9336618	-0.3072449	0.1164942

Individual P-values (lower = worse):

```

    edges b1degree1    cycle4
0.3504784 0.7586570 0.9072609
Joint P-value (lower = worse): 0.1702401
Chain 2

```

```

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```

```

    edges    b1degree1    cycle4
0.009917489 -0.849595199 0.751238982

```

```

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.9920871 0.3955502 0.4525088
Joint P-value (lower = worse): 0.4049945
Chain 3

```

```

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```

```

    edges b1degree1    cycle4
-0.2985873 0.6818755 -1.1698052

```

```

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.7652550 0.4953177 0.2420794
Joint P-value (lower = worse): 0.5700744
Chain 4

```

```

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```

```

    edges b1degree1    cycle4
0.1115741 -0.9752093 0.1768475

```

```

Individual P-values (lower = worse):
    edges b1degree1    cycle4
0.9111611 0.3294564 0.8596282
Joint P-value (lower = worse): 0.5713822
Chain 5

```

```

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```

```

    edges b1degree1    cycle4
-0.1818034 0.3848732 -0.4954972

```

Individual P-values (lower = worse):
edges b1degree1 cycle4
0.8557370 0.7003313 0.6202492
Joint P-value (lower = worse): 0.8975539
Chain 6

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges b1degree1 cycle4
-0.4751854 0.8079716 -0.2877042

Individual P-values (lower = worse):
edges b1degree1 cycle4
0.6346548 0.4191069 0.7735732
Joint P-value (lower = worse): 0.6088046
Chain 7

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges b1degree1 cycle4
-0.8753117 -0.1067233 -0.7792244

Individual P-values (lower = worse):
edges b1degree1 cycle4
0.3814043 0.9150085 0.4358475
Joint P-value (lower = worse): 0.5840519
Chain 8

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

edges b1degree1 cycle4
-0.4947983 1.4777652 -1.9407711

Individual P-values (lower = worse):
edges b1degree1 cycle4
0.62074253 0.13947063 0.05228605
Joint P-value (lower = worse): 0.1057777
Chain 9

Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

```

edges b1degree1 cycle4
0.7249734 0.1051507 0.8464796

```

Individual P-values (lower = worse):

```

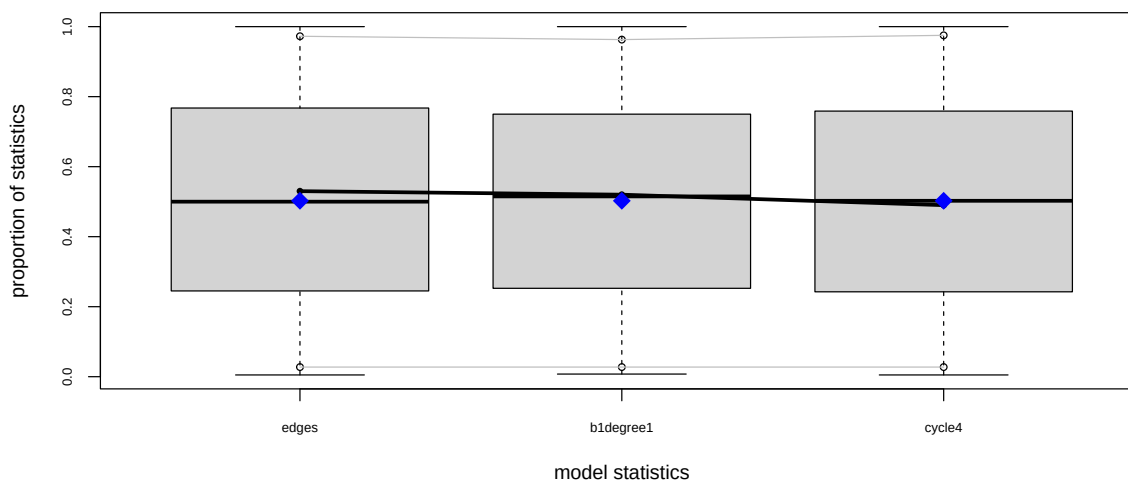
edges b1degree1 cycle4
0.4684684 0.9162563 0.3972853

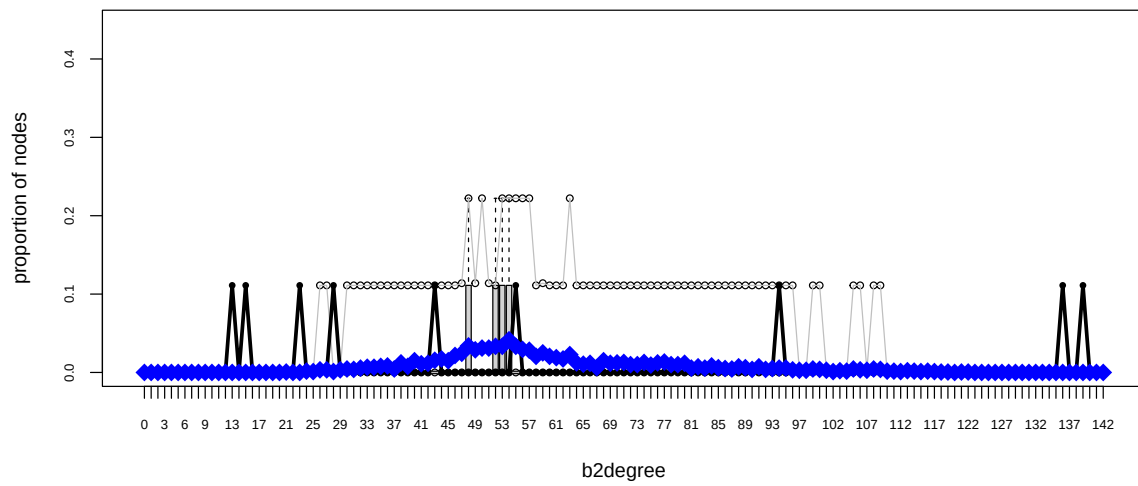
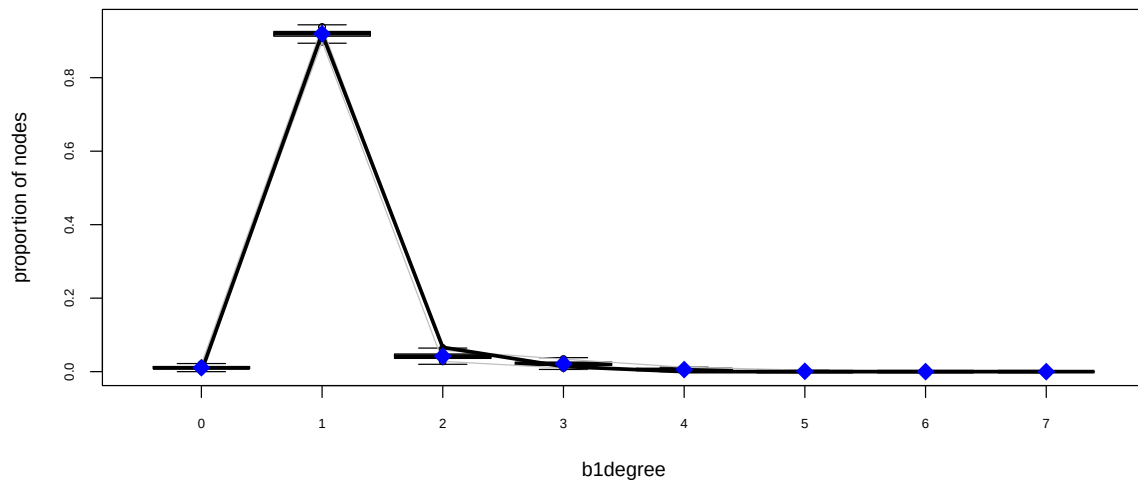
```

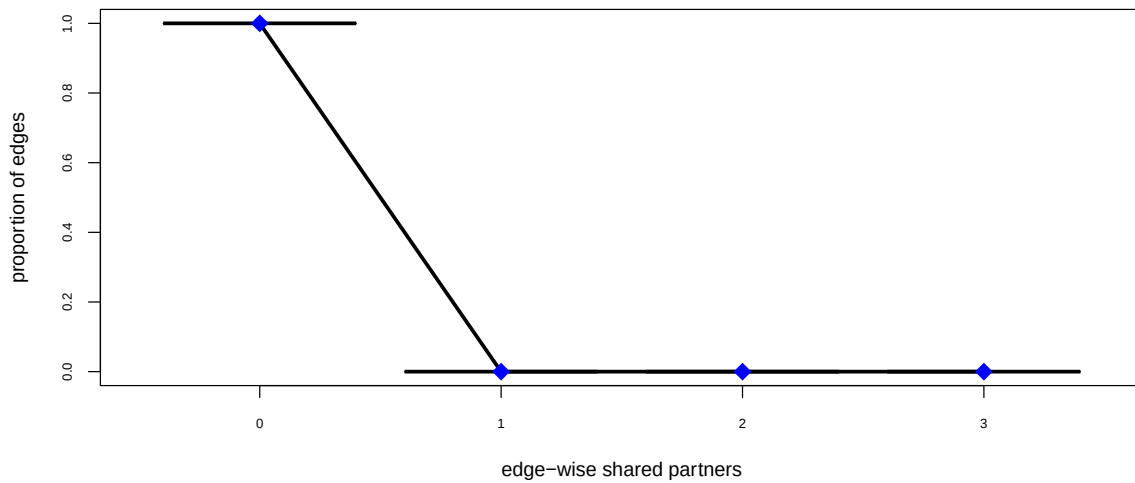
Joint P-value (lower = worse): 0.725172

Note: To save space, only one in every 4 iterations of the MCMC sample used for estimation was stored for diagnostics. Sample size per chain was originally around 15860 with thinning interval 16.

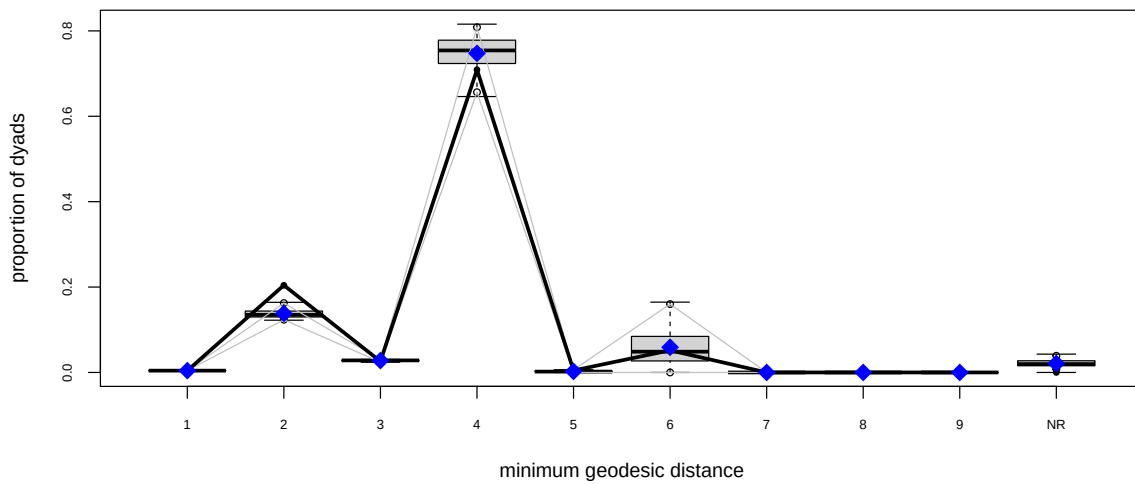
Note: MCMC diagnostics shown here are from the last round of simulation, prior to computation of final parameter estimates. Because the final estimates are refinements of those used for this simulation run, these diagnostics may understate model performance. To directly assess the performance of the final model on in-model statistics, please use the GOF command: `gof(ergmFitObject, GOF=~model)`.







Goodness-of-fit diagnostics



Goodness-of-fit for model statistics

	obs	min	mean	max	MC	p-value
edges	546	528	546.330	568		1.00
bldegree1	459	446	458.785	473		1.00
cycle4	67	26	68.885	157		0.98

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max	MC	p-value
1	546	528	546.330	568		1.00

2	26301	15759	17794.830	21669	0.00
3	3351	2845	3587.910	3920	0.33
4	91309	77582	96233.180	105053	0.37
5	594	0	308.520	1061	0.25
6	6677	0	7603.865	28137	0.96
Inf	0	0	2703.365	6018	0.01

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max	MC	p-value
0	0	0	5.360	12		0.01
1	459	446	458.785	473		1.00
2	33	10	20.880	32		0.00
3	7	3	10.670	21		0.30
4	0	0	2.790	7		0.10
5	0	0	0.475	3		1.00
6	0	0	0.040	2		1.00

Goodness-of-fit for bipartition 2 degree

	obs	min	mean	max	MC	p-value
13	1	0	0.000	0		0.00
15	1	0	0.000	0		0.00
21	0	0	0.005	1		1.00
23	1	0	0.000	0		0.00
24	0	0	0.015	2		1.00
25	0	0	0.010	1		1.00
26	0	0	0.030	1		1.00
27	0	0	0.030	1		1.00
28	1	0	0.010	1		0.02
29	0	0	0.030	2		1.00
30	0	0	0.040	2		1.00
31	0	0	0.035	1		1.00
32	0	0	0.050	2		1.00
33	0	0	0.060	1		1.00
34	0	0	0.060	1		1.00
35	0	0	0.070	2		1.00
36	0	0	0.075	1		1.00
37	0	0	0.040	1		1.00
38	0	0	0.110	1		1.00
39	0	0	0.070	2		1.00
40	0	0	0.135	1		1.00
41	0	0	0.100	3		1.00
42	0	0	0.105	3		1.00
43	1	0	0.140	2		0.27
44	0	0	0.155	2		1.00

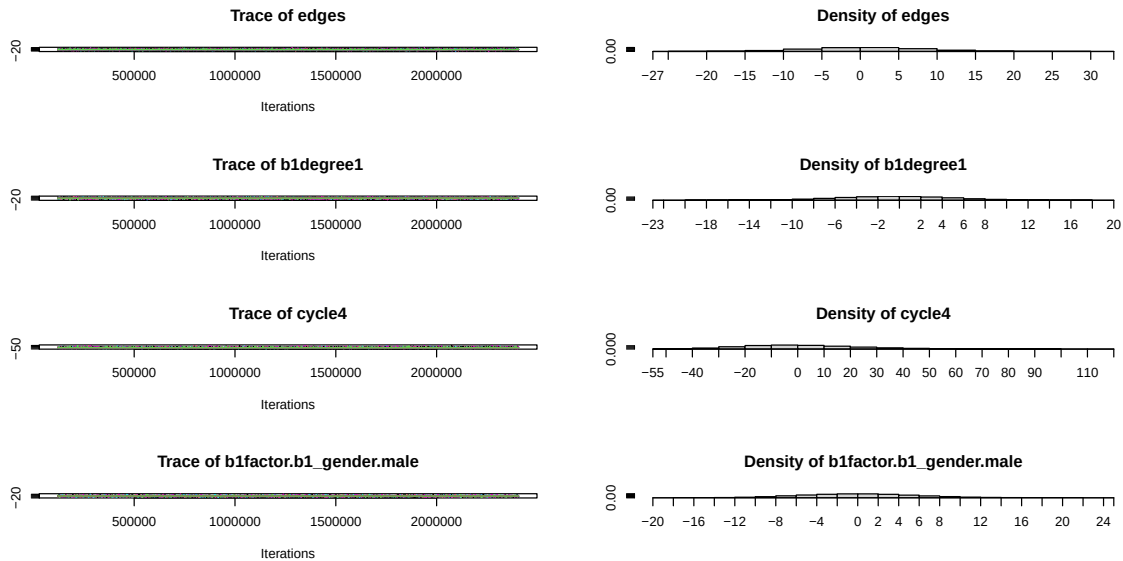
45	0	0	0.135	2	1.00
46	0	0	0.195	2	1.00
47	0	0	0.220	2	1.00
48	0	0	0.305	2	1.00
49	0	0	0.260	3	1.00
50	0	0	0.280	3	1.00
51	0	0	0.280	3	1.00
52	0	0	0.300	2	1.00
53	0	0	0.300	2	1.00
54	0	0	0.375	2	1.00
55	1	0	0.300	3	0.48
56	0	0	0.265	4	1.00
57	0	0	0.255	2	1.00
58	0	0	0.185	3	1.00
59	0	0	0.225	2	1.00
60	0	0	0.185	3	1.00
61	0	0	0.170	2	1.00
62	0	0	0.155	3	1.00
63	0	0	0.205	3	1.00
64	0	0	0.110	2	1.00
65	0	0	0.095	1	1.00
66	0	0	0.105	2	1.00
67	0	0	0.055	2	1.00
68	0	0	0.135	2	1.00
69	0	0	0.105	2	1.00
70	0	0	0.110	2	1.00
71	0	0	0.115	2	1.00
72	0	0	0.090	2	1.00
73	0	0	0.090	2	1.00
74	0	0	0.115	2	1.00
75	0	0	0.085	2	1.00
76	0	0	0.110	2	1.00
77	0	0	0.120	1	1.00
78	0	0	0.090	2	1.00
79	0	0	0.090	2	1.00
80	0	0	0.105	2	1.00
81	0	0	0.050	1	1.00
82	0	0	0.065	1	1.00
83	0	0	0.050	1	1.00
84	0	0	0.075	1	1.00
85	0	0	0.055	1	1.00
86	0	0	0.045	1	1.00
87	0	0	0.040	2	1.00
88	0	0	0.065	2	1.00
89	0	0	0.055	1	1.00
90	0	0	0.035	1	1.00

91	0	0	0.065	1	1.00
92	0	0	0.035	2	1.00
93	0	0	0.045	1	1.00
94	1	0	0.050	2	0.09
95	0	0	0.050	1	1.00
96	0	0	0.030	1	1.00
97	0	0	0.025	1	1.00
98	0	0	0.025	1	1.00
99	0	0	0.040	1	1.00
100	0	0	0.035	1	1.00
101	0	0	0.025	1	1.00
102	0	0	0.010	1	1.00
103	0	0	0.020	1	1.00
104	0	0	0.015	1	1.00
105	0	0	0.040	1	1.00
106	0	0	0.030	1	1.00
107	0	0	0.025	1	1.00
108	0	0	0.040	1	1.00
109	0	0	0.035	1	1.00
110	0	0	0.015	1	1.00
111	0	0	0.015	1	1.00
112	0	0	0.010	1	1.00
113	0	0	0.020	1	1.00
114	0	0	0.015	1	1.00
115	0	0	0.010	1	1.00
116	0	0	0.015	1	1.00
117	0	0	0.010	1	1.00
118	0	0	0.005	1	1.00
119	0	0	0.005	1	1.00
121	0	0	0.005	1	1.00
122	0	0	0.005	1	1.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max MC	p-value
546.00	528.00	546.33	568.00	1.00

A.4.4 Model 3



Sample statistics summary:

Iterations = 120448:2406528
 Thinning interval = 640
 Number of chains = 9
 Sample size per chain = 3573

1. Empirical mean and standard deviation for each variable,
 plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
edges	1.25307	7.431	0.04144	0.06826
b1degree1	-0.05458	5.252	0.02929	0.04472
cycle4	2.61648	22.356	0.12467	0.18772
b1factor.b1_gender.male	0.71207	5.713	0.03186	0.05854

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
edges	-13	-4	1	6	16
b1degree1	-11	-4	0	4	10
cycle4	-34	-13	0	16	52
b1factor.b1_gender.male	-10	-3	1	4	12

Are sample statistics significantly different from observed?

edges b1degree1 cycle4 b1factor.b1_gender.male

diff.	1.253071e+00	-0.05457599	2.616475e+00	7.120689e-01
test stat.	1.837232e+01	-1.22226000	1.394755e+01	1.216653e+01
P-val.	2.188414e-75	0.22160932	3.256092e-44	4.685805e-34

(Omni)

diff.	NA
test stat.	4.981756e+02
P-val.	2.180719e-104

Sample statistics cross-correlations:

	edges	b1degree1	cycle4
edges	1.0000000	-0.6273622	0.8549354
b1degree1	-0.6273622	1.0000000	-0.5698004
cycle4	0.8549354	-0.5698004	1.0000000
bifactor.b1_gender.male	0.5314239	-0.3247677	0.4217988

	bifactor.b1_gender.male
edges	0.5314239
b1degree1	-0.3247677
cycle4	0.4217988
bifactor.b1_gender.male	1.0000000

Sample statistics auto-correlation:

Chain 1

	edges	b1degree1	cycle4	bifactor.b1_gender.male
Lag 0	1.00000000	1.00000000	1.00000000	1.00000000
Lag 640	0.46219220	0.36123841	0.36808454	0.56555488
Lag 1280	0.22083629	0.13173911	0.15681895	0.32971930
Lag 1920	0.10106233	0.05114607	0.06465253	0.19505666
Lag 2560	0.04911140	0.02504884	0.03203135	0.10243262
Lag 3200	0.01765148	-0.01676550	0.03047180	0.04651129

Chain 2

	edges	b1degree1	cycle4	bifactor.b1_gender.male
Lag 0	1.000000000	1.00000000	1.00000000	1.00000000
Lag 640	0.437858208	0.38674645	0.35549994	0.54439105
Lag 1280	0.170365470	0.17315370	0.11391348	0.30861060
Lag 1920	0.077068682	0.07864354	0.04139655	0.19067311
Lag 2560	0.014848922	0.03272962	0.01527084	0.11125867
Lag 3200	-0.002735924	0.02799260	0.01029926	0.06692818

Chain 3

	edges	b1degree1	cycle4	bifactor.b1_gender.male
Lag 0	1.0000000000	1.00000000	1.00000000	1.00000000
Lag 640	0.4638303468	0.38499285	0.36101793	0.56588893
Lag 1280	0.2486813189	0.18855567	0.18681347	0.33965285
Lag 1920	0.1317563283	0.10151423	0.09475260	0.19857586
Lag 2560	0.0475566842	0.05226031	0.04985226	0.09518536
Lag 3200	0.0003148158	0.02992456	0.02826998	0.01726816

Chain 4

	edges	b1degree1	cycle4	b1factor.b1_gender.male
Lag 0	1.000000e+00	1.00000000	1.00000000	1.00000000
Lag 640	4.474375e-01	0.40827558	0.36602612	0.56673910
Lag 1280	1.999899e-01	0.18376229	0.16108594	0.33121846
Lag 1920	7.810462e-02	0.06819393	0.06662650	0.17515468
Lag 2560	2.214036e-02	0.03229893	0.02759365	0.07527084
Lag 3200	1.327434e-05	0.03020447	0.01418929	0.04068117

Chain 5

	edges	b1degree1	cycle4	b1factor.b1_gender.male
Lag 0	1.000000000	1.00000000	1.00000000	1.00000000
Lag 640	0.437103052	0.38423058	0.34784445	0.56856874
Lag 1280	0.216170483	0.17224360	0.15877064	0.33285961
Lag 1920	0.105458281	0.07302384	0.09177188	0.19717275
Lag 2560	0.044444789	0.02504269	0.04868227	0.08931366
Lag 3200	-0.009955747	-0.01250088	-0.01686271	0.04036875

Chain 6

	edges	b1degree1	cycle4	b1factor.b1_gender.male
Lag 0	1.000000000	1.000000000	1.00000000	1.00000000
Lag 640	0.45435011	0.361268517	0.35812343	0.56960682
Lag 1280	0.21566779	0.153981676	0.15965314	0.33619428
Lag 1920	0.12654181	0.072802022	0.09339018	0.18847000
Lag 2560	0.05006227	0.007440694	0.03046906	0.09862146
Lag 3200	0.02497662	-0.008779587	0.01561748	0.02754053

Chain 7

	edges	b1degree1	cycle4	b1factor.b1_gender.male
Lag 0	1.000000000	1.000000000	1.000000000	1.00000000
Lag 640	0.463362102	0.362904627	0.363305483	0.56274889
Lag 1280	0.207687456	0.144460485	0.156659826	0.30735588
Lag 1920	0.089675932	0.044654202	0.078071052	0.17663718
Lag 2560	0.036751121	0.023897805	0.035773901	0.10232906
Lag 3200	0.003172149	0.006837233	-0.001948617	0.07518207

Chain 8

	edges	b1degree1	cycle4	b1factor.b1_gender.male
Lag 0	1.000000000	1.000000000	1.000000000	1.00000000
Lag 640	0.44705209	0.40603636	0.341088943	0.55333751
Lag 1280	0.22469509	0.18881850	0.160573033	0.31794405
Lag 1920	0.10221498	0.08973941	0.058634998	0.19089898
Lag 2560	0.03224681	0.04772215	0.027979386	0.10318507
Lag 3200	0.01803967	0.02104698	0.001062492	0.05033198

Chain 9

	edges	b1degree1	cycle4	b1factor.b1_gender.male
Lag 0	1.000000000	1.000000000	1.000000000	1.00000000
Lag 640	0.44119476	0.365874321	0.343237943	0.54715732
Lag 1280	0.21378712	0.165901421	0.149457193	0.29771860
Lag 1920	0.09606213	0.056225445	0.054445652	0.15065216
Lag 2560	0.03676788	0.018721924	-0.002869194	0.06468348

Lag 3200 0.03169008 0.002035173 0.007041516 0.02444377

Sample statistics burn-in diagnostic (Geweke):

Chain 1

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-0.8376964	-0.4855318	-0.5127717
bifactor.b1_gender.male	-0.3354309		

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.4022012	0.6272992	0.6081110
bifactor.b1_gender.male	0.7373001		

Joint P-value (lower = worse): 0.8182054

Chain 2

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-2.2283677	-0.1273937	-1.7492323
bifactor.b1_gender.male	-1.1146404		

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.02585601	0.89862881	0.08025087
bifactor.b1_gender.male	0.26500453		

Joint P-value (lower = worse): 0.09741188

Chain 3

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	0.12534550	-0.54659601	-0.09595021
bifactor.b1_gender.male	-0.46754728		

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.9002500	0.5846563	0.9235601
bifactor.b1_gender.male	0.6401084		

Joint P-value (lower = worse): 0.8205227

Chain 4

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-0.8731893	0.4147736	-0.6828871
bifactor.b1_gender.male	-1.0626114		

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.3825599	0.6783077	0.4946782
bifactor.b1_gender.male	0.2879582		

Joint P-value (lower = worse): 0.8571621

Chain 5

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-0.7303596	0.4544881	-0.7449774
bifactor.b1_gender.male	-0.7221944		

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.4651704	0.6494776	0.4562854
bifactor.b1_gender.male	0.4701749		

Joint P-value (lower = worse): 0.9284842

Chain 6

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-0.4456012	-0.3455419	-0.7847546
bifactor.b1_gender.male	-0.2160909		

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.6558853	0.7296870	0.4325975
bifactor.b1_gender.male	0.8289169		

Joint P-value (lower = worse): 0.7642974

Chain 7

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-0.3819243	0.7027031	-0.7415056
bifactor.b1_gender.male	1.4539718		

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.7025175	0.4822408	0.4583870
bifactor.b1_gender.male	0.1459541		

Joint P-value (lower = worse): 0.5891684

Chain 8

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-0.104999617	-0.968118379	-0.004073132
bifactor.b1_gender.male	-0.599161923		

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.9163761	0.3329853	0.9967501
bifactor.b1_gender.male	0.5490649		

Joint P-value (lower = worse): 0.6925802

Chain 9

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	1.2922259	-0.9945804	1.0602979

```
bifactor.b1_gender.male
1.4268769
```

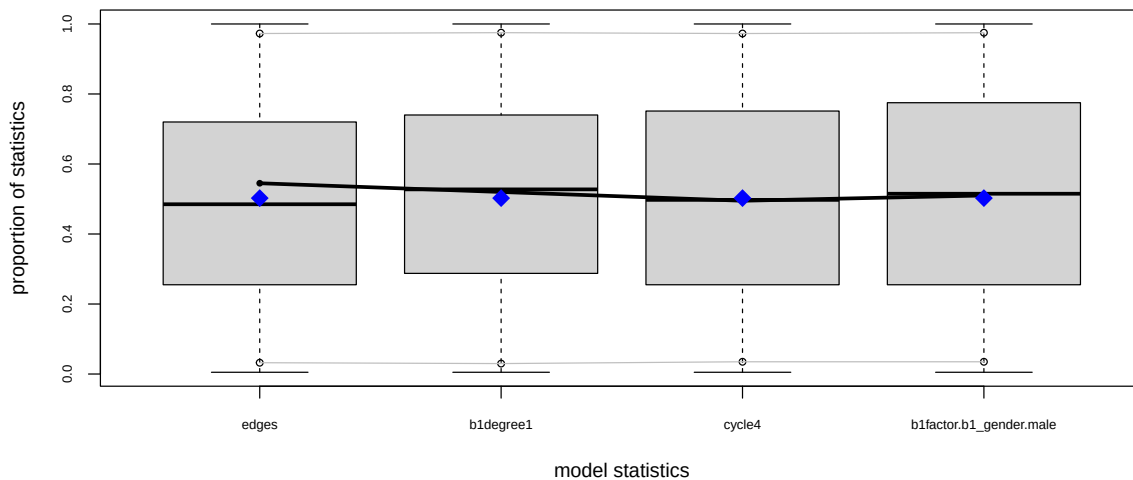
Individual P-values (lower = worse):

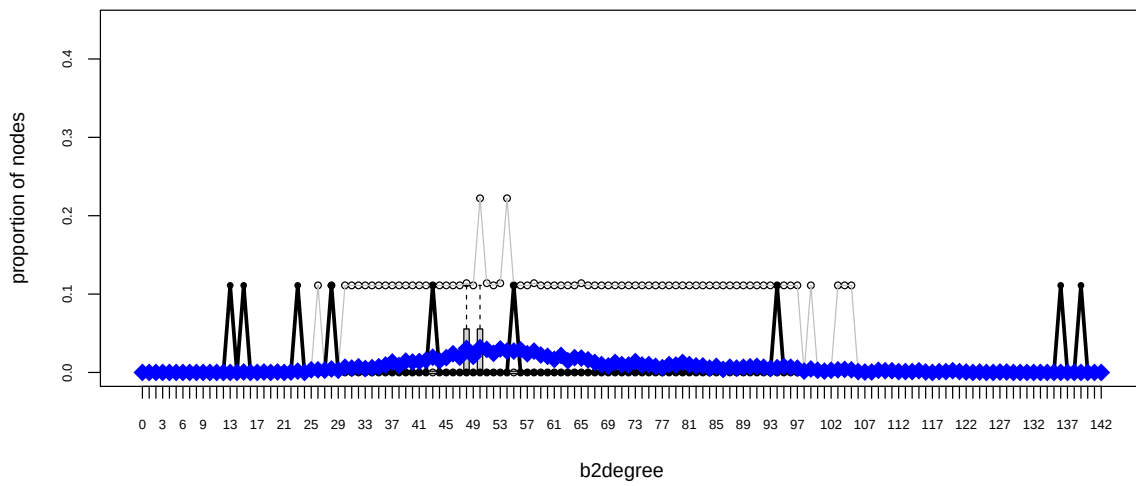
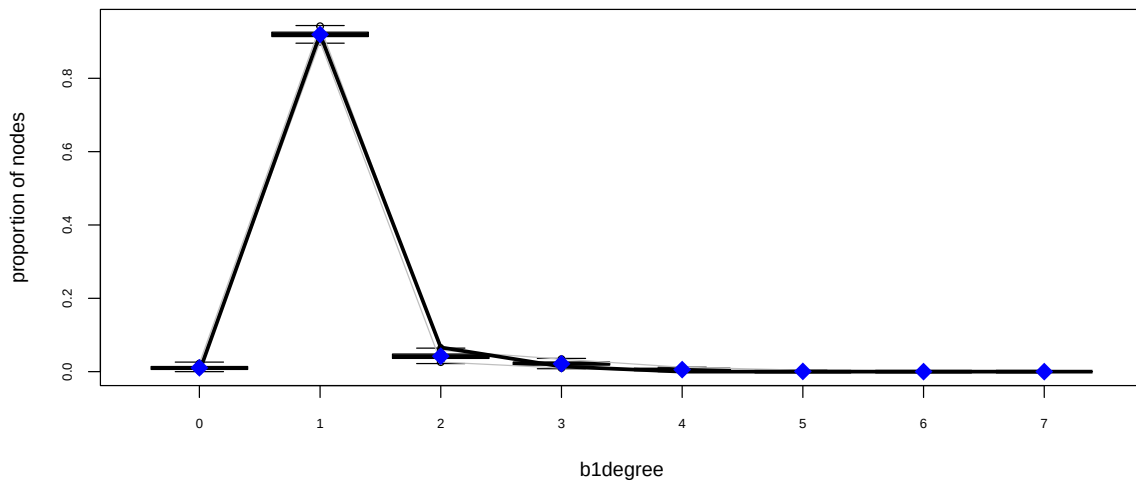
	edges	b1degree1	cycle4
	0.1962789	0.3199404	0.2890091
bifactor.b1_gender.male			
	0.1536154		

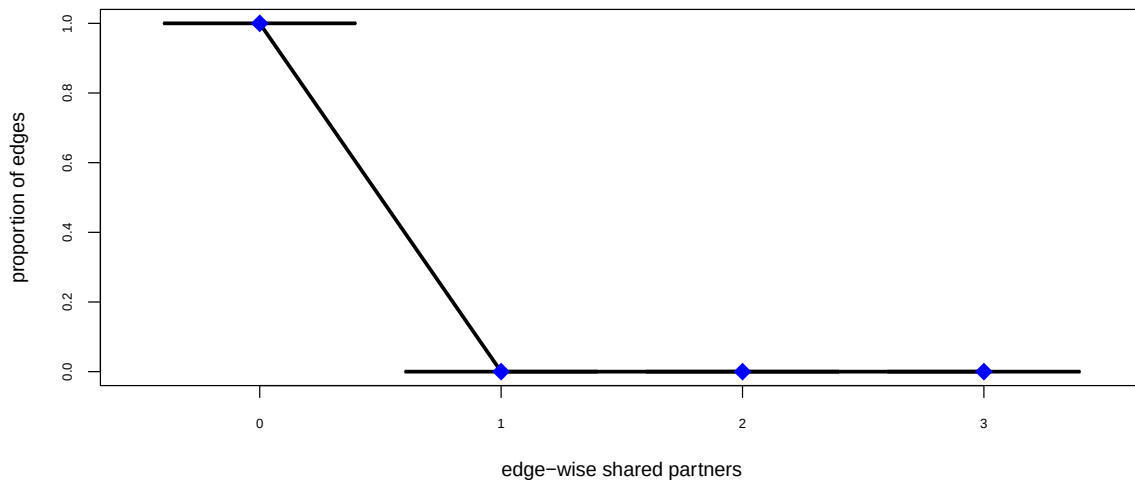
Joint P-value (lower = worse): 0.5098452

Note: To save space, only one in every 5 iterations of the MCMC sample used for estimation was stored for diagnostics. Sample size per chain was originally around 17865 with thinning interval 128.

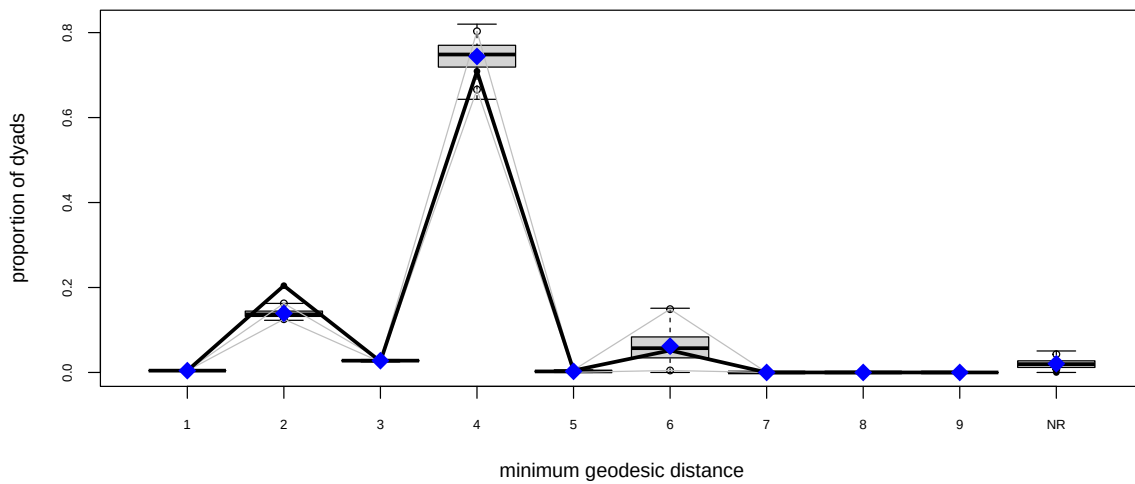
Note: MCMC diagnostics shown here are from the last round of simulation, prior to computation of final parameter estimates. Because the final estimates are refinements of those used for this simulation run, these diagnostics may understate model performance. To directly assess the performance of the final model on in-model statistics, please use the GOF command: `gof(ergmFitObject, GOF=~model)`.







Goodness-of-fit diagnostics



Goodness-of-fit for model statistics

	obs	min	mean	max	MC	p-value
edges	546	527	546.070	570		1.00
b1degree1	459	441	458.810	474		1.00
cycle4	67	20	67.250	144		0.99
b1factor.b1_gender.male	257	243	256.995	274		1.00

Goodness-of-fit for minimum geodesic distance

obs	min	mean	max	MC	p-value
-----	-----	------	-----	----	---------

1	546	527	546.070	570	1.00
2	26301	15809	17922.080	22664	0.00
3	3351	2921	3563.925	3912	0.26
4	91309	79282	95788.770	105587	0.34
5	594	0	333.755	953	0.18
6	6677	0	7976.075	24134	0.92
Inf	0	0	2647.325	6513	0.01

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max	MC	p-value
0	0	0	5.250	13		0.01
1	459	441	458.810	474		1.00
2	33	9	21.095	32		0.00
3	7	3	10.715	18		0.30
4	0	0	2.755	7		0.13
5	0	0	0.345	2		1.00
6	0	0	0.030	1		1.00

Goodness-of-fit for bipartition 2 degree

	obs	min	mean	max	MC	p-value
13	1	0	0.000	0		0.00
15	1	0	0.005	1		0.01
18	0	0	0.005	1		1.00
20	0	0	0.005	1		1.00
22	0	0	0.005	1		1.00
23	1	0	0.015	1		0.03
25	0	0	0.030	2		1.00
26	0	0	0.030	1		1.00
27	0	0	0.025	1		1.00
28	1	0	0.040	2		0.07
29	0	0	0.025	1		1.00
30	0	0	0.060	1		1.00
31	0	0	0.050	1		1.00
32	0	0	0.065	2		1.00
33	0	0	0.045	1		1.00
34	0	0	0.055	1		1.00
35	0	0	0.065	1		1.00
36	0	0	0.085	2		1.00
37	0	0	0.120	2		1.00
38	0	0	0.080	2		1.00
39	0	0	0.130	2		1.00
40	0	0	0.120	2		1.00
41	0	0	0.125	2		1.00
42	0	0	0.140	1		1.00

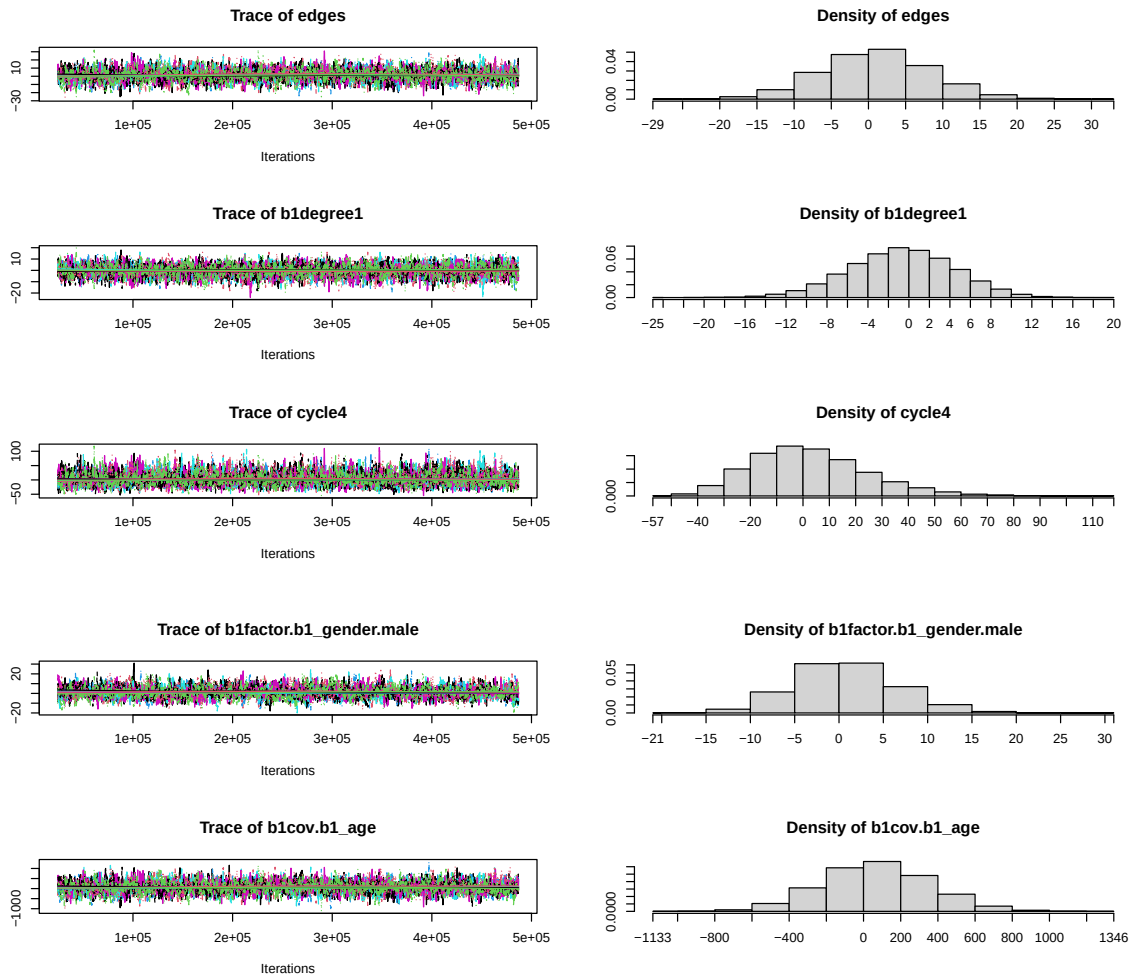
43	1	0	0.175	2	0.33
44	0	0	0.135	1	1.00
45	0	0	0.170	2	1.00
46	0	0	0.215	3	1.00
47	0	0	0.175	2	1.00
48	0	0	0.275	2	1.00
49	0	0	0.185	2	1.00
50	0	0	0.285	2	1.00
51	0	0	0.265	2	1.00
52	0	0	0.220	2	1.00
53	0	0	0.270	2	1.00
54	0	0	0.245	4	1.00
55	1	0	0.245	2	0.46
56	0	0	0.260	2	1.00
57	0	0	0.215	2	1.00
58	0	0	0.245	2	1.00
59	0	0	0.200	2	1.00
60	0	0	0.185	2	1.00
61	0	0	0.150	2	1.00
62	0	0	0.195	2	1.00
63	0	0	0.135	2	1.00
64	0	0	0.170	2	1.00
65	0	0	0.165	2	1.00
66	0	0	0.145	2	1.00
67	0	0	0.115	1	1.00
68	0	0	0.090	2	1.00
69	0	0	0.075	1	1.00
70	0	0	0.115	2	1.00
71	0	0	0.090	2	1.00
72	0	0	0.085	1	1.00
73	0	0	0.125	1	1.00
74	0	0	0.085	1	1.00
75	0	0	0.090	2	1.00
76	0	0	0.070	1	1.00
77	0	0	0.055	1	1.00
78	0	0	0.090	2	1.00
79	0	0	0.085	1	1.00
80	0	0	0.115	2	1.00
81	0	0	0.090	1	1.00
82	0	0	0.075	1	1.00
83	0	0	0.075	1	1.00
84	0	0	0.050	1	1.00
85	0	0	0.070	1	1.00
86	0	0	0.035	1	1.00
87	0	0	0.060	1	1.00
88	0	0	0.050	1	1.00

89	0	0	0.060	1	1.00
90	0	0	0.065	1	1.00
91	0	0	0.070	2	1.00
92	0	0	0.060	1	1.00
93	0	0	0.045	2	1.00
94	1	0	0.055	1	0.11
95	0	0	0.070	1	1.00
96	0	0	0.060	1	1.00
97	0	0	0.050	1	1.00
98	0	0	0.015	1	1.00
99	0	0	0.040	1	1.00
100	0	0	0.025	1	1.00
101	0	0	0.015	1	1.00
102	0	0	0.025	1	1.00
103	0	0	0.030	1	1.00
104	0	0	0.035	1	1.00
105	0	0	0.030	1	1.00
106	0	0	0.010	1	1.00
107	0	0	0.005	1	1.00
108	0	0	0.005	1	1.00
109	0	0	0.025	1	1.00
110	0	0	0.020	1	1.00
111	0	0	0.020	1	1.00
112	0	0	0.010	1	1.00
113	0	0	0.010	1	1.00
114	0	0	0.010	1	1.00
115	0	0	0.020	1	1.00
116	0	0	0.005	1	1.00
118	0	0	0.010	1	1.00
119	0	0	0.010	1	1.00
120	0	0	0.020	1	1.00
121	0	0	0.010	1	1.00
122	0	0	0.005	1	1.00
124	0	0	0.005	1	1.00
127	0	0	0.005	1	1.00
128	0	0	0.005	1	1.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max MC p-value
546.00	527.00	546.07	570.00

A.4.5 Model 4



Sample statistics summary:

Iterations = 24397:487041
 Thinning interval = 124
 Number of chains = 9
 Sample size per chain = 3732

1. Empirical mean and standard deviation for each variable,
 plus standard error of the mean:

Mean	SD	Naive SE	Time-series SE
------	----	----------	----------------

edges	1.4885	7.342	0.04006	0.13991
b1degree1	-0.2421	5.158	0.02815	0.08707
cycle4	2.9974	21.952	0.11978	0.36758
b1factor.b1_gender.male	1.2371	5.804	0.03167	0.13091
b1cov.b1_age	68.2142	290.100	1.58291	5.53954

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
edges	-13.0	-3	2	6	16
b1degree1	-11.0	-4	0	3	9
cycle4	-34.0	-13	1	16	52
b1factor.b1_gender.male	-10.0	-3	1	5	13
b1cov.b1_age	-493.3	-128	66	262	642

Are sample statistics significantly different from observed?

	edges	b1degree1	cycle4	b1factor.b1_gender.male
diff.	1.488508e+00	-0.242080505	2.997410e+00	1.237108e+00
test stat.	1.067485e+01	-2.788939985	8.253381e+00	9.459740e+00
P-val.	1.334799e-26	0.005288086	1.539758e-16	3.087086e-21

	b1cov.b1_age	(Omni)
diff.	6.821421e+01	NA
test stat.	1.234136e+01	1.836478e+02
P-val.	5.423515e-35	1.961837e-36

Sample statistics cross-correlations:

	edges	b1degree1	cycle4
edges	1.0000000	-0.6194906	0.8444657
b1degree1	-0.6194906	1.0000000	-0.5563244
cycle4	0.8444657	-0.5563244	1.0000000
b1factor.b1_gender.male	0.5202118	-0.3085227	0.4001146
b1cov.b1_age	0.9278442	-0.5648680	0.7655343

	b1factor.b1_gender.male	b1cov.b1_age
edges	0.5202118	0.9278442
b1degree1	-0.3085227	-0.5648680
cycle4	0.4001146	0.7655343
b1factor.b1_gender.male	1.0000000	0.4940175
b1cov.b1_age	0.4940175	1.0000000

Sample statistics auto-correlation:

Chain 1

	edges	b1degree1	cycle4	b1factor.b1_gender.male	b1cov.b1_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8374849	0.7942875	0.7803126	0.8871185	0.8414362
Lag 248	0.6983437	0.6303555	0.6121622	0.7828993	0.7078650

Lag 372	0.5875911	0.5139794	0.4829741	0.6943681	0.5983029
Lag 496	0.4942324	0.4232287	0.3869254	0.6236215	0.5049571
Lag 620	0.4165554	0.3530637	0.3131236	0.5611651	0.4286563

Chain 2

	edges	b1degree1	cycle4	bifactor.b1_gender.male	b1cov.b1_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8383225	0.7916470	0.7806649	0.8891942	0.8368086
Lag 248	0.7002352	0.6310644	0.6246221	0.7854262	0.6984174
Lag 372	0.5905196	0.5020604	0.5019041	0.6964845	0.5904724
Lag 496	0.5001746	0.4028386	0.4089298	0.6196010	0.5006779
Lag 620	0.4244166	0.3227622	0.3352208	0.5524969	0.4271361

Chain 3

	edges	b1degree1	cycle4	bifactor.b1_gender.male	b1cov.b1_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8615095	0.8196269	0.7953017	0.8950554	0.8572147
Lag 248	0.7410270	0.6783742	0.6389925	0.8019235	0.7339498
Lag 372	0.6323685	0.5562304	0.5143062	0.7177998	0.6210777
Lag 496	0.5423396	0.4579825	0.4248686	0.6461454	0.5253416
Lag 620	0.4711679	0.3790521	0.3656032	0.5799580	0.4538163

Chain 4

	edges	b1degree1	cycle4	bifactor.b1_gender.male	b1cov.b1_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8561850	0.8062786	0.7835227	0.8972264	0.8625570
Lag 248	0.7406347	0.6512770	0.6281802	0.8069659	0.7508933
Lag 372	0.6429638	0.5281485	0.5154130	0.7245146	0.6543449
Lag 496	0.5597939	0.4335170	0.4229578	0.6536452	0.5722606
Lag 620	0.4859649	0.3583084	0.3431372	0.5874509	0.4985239

Chain 5

	edges	b1degree1	cycle4	bifactor.b1_gender.male	b1cov.b1_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8483661	0.8106589	0.7941002	0.8895643	0.8523591
Lag 248	0.7166964	0.6615552	0.6418816	0.7910443	0.7234805
Lag 372	0.6070666	0.5521223	0.5189634	0.6987463	0.6181309
Lag 496	0.5112907	0.4615184	0.4173753	0.6120248	0.5242244
Lag 620	0.4308833	0.3952802	0.3465701	0.5350108	0.4454119

Chain 6

	edges	b1degree1	cycle4	bifactor.b1_gender.male	b1cov.b1_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8440858	0.8169353	0.7926372	0.8794121	0.8483472
Lag 248	0.7072567	0.6661694	0.6289095	0.7780482	0.7142171
Lag 372	0.5889945	0.5478361	0.5072255	0.6925403	0.5967909
Lag 496	0.4919709	0.4477178	0.4088420	0.6202981	0.4991670
Lag 620	0.4093479	0.3623102	0.3314522	0.5515389	0.4146804

Chain 7

	edges	b1degree1	cycle4	bifactor.b1_gender.male	b1cov.b1_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000

Lag 124	0.8276469	0.7840563	0.7680361	0.8812961	0.8303218
Lag 248	0.6838055	0.6143846	0.5907489	0.7776281	0.6871530
Lag 372	0.5563137	0.4767232	0.4611630	0.6858163	0.5627388
Lag 496	0.4511159	0.3735897	0.3588736	0.6088053	0.4610083
Lag 620	0.3611492	0.2954504	0.2796229	0.5461082	0.3715808

Chain 8

	edges	b1degree1	cycle4	bifactor.b1_gender.male	b1cov.b1_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8502719	0.8158044	0.8033231	0.8961069	0.8551644
Lag 248	0.7212653	0.6720300	0.6556043	0.8067487	0.7284289
Lag 372	0.6089928	0.5499170	0.5400857	0.7258174	0.6171646
Lag 496	0.5151437	0.4553121	0.4498942	0.6572668	0.5236445
Lag 620	0.4360382	0.3809871	0.3782412	0.5912337	0.4455866

Chain 9

	edges	b1degree1	cycle4	bifactor.b1_gender.male	b1cov.b1_age
Lag 0	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Lag 124	0.8566705	0.8092882	0.7907467	0.8862144	0.8512702
Lag 248	0.7370720	0.6654765	0.6403479	0.7898657	0.7272172
Lag 372	0.6361992	0.5476526	0.5251665	0.7077886	0.6240128
Lag 496	0.5434866	0.4496682	0.4222984	0.6302241	0.5327234
Lag 620	0.4643092	0.3647572	0.3475467	0.5626955	0.4569503

Sample statistics burn-in diagnostic (Geweke):

Chain 1

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-0.1688130	-0.3143331	0.1829230
bifactor.b1_gender.male		b1cov.b1_age	
	0.5842642	0.3253693	

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.8659437	0.7532680	0.8548584
bifactor.b1_gender.male		b1cov.b1_age	
	0.5590426	0.7449016	

Joint P-value (lower = worse): 0.8466799

Chain 2

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	0.1634696	-1.7034344	-0.1104156

bifactor.b1_gender.male	b1cov.b1_age
-1.3277024	-0.3132087

Individual P-values (lower = worse):

edges	b1degree1	cycle4
0.87014867	0.08848681	0.91207977
bifactor.b1_gender.male	b1cov.b1_age	
0.18427644	0.75412209	

Joint P-value (lower = worse): 0.3843382

Chain 3

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

edges	b1degree1	cycle4
-2.7773335	0.7191514	-2.9640630
bifactor.b1_gender.male	b1cov.b1_age	
-0.2985897	-1.4002474	

Individual P-values (lower = worse):

edges	b1degree1	cycle4
0.005480690	0.472047664	0.003036062
bifactor.b1_gender.male	b1cov.b1_age	
0.765253095	0.161439250	

Joint P-value (lower = worse): 0.02025937

Chain 4

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

edges	b1degree1	cycle4
-0.9135621	0.9209371	-1.9148485
bifactor.b1_gender.male	b1cov.b1_age	
2.0238862	-0.7310445	

Individual P-values (lower = worse):

edges	b1degree1	cycle4
0.36094699	0.35708327	0.05551182
bifactor.b1_gender.male	b1cov.b1_age	
0.04298186	0.46475196	

Joint P-value (lower = worse): 0.1181432

Chain 5

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

edges	b1degree1	cycle4
-0.83100175	2.14295138	-1.38371681
bifactor.b1_gender.male	b1cov.b1_age	
0.91877545	-0.06611033	

Individual P-values (lower = worse):

edges	b1degree1	cycle4
0.4059726	0.0321170	0.1664452
bifactor.b1_gender.male	b1cov.b1_age	
0.3582130	0.9472900	

Joint P-value (lower = worse): 0.4623169

Chain 6

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

edges	b1degree1	cycle4
-0.81548604	0.71614808	-1.40917569
bifactor.b1_gender.male	b1cov.b1_age	
0.71236910	0.07078785	

Individual P-values (lower = worse):

edges	b1degree1	cycle4
0.4147942	0.4738999	0.1587832
bifactor.b1_gender.male	b1cov.b1_age	
0.4762362	0.9435666	

Joint P-value (lower = worse): 0.4325175

Chain 7

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

edges	b1degree1	cycle4
0.5264058	-0.9639856	0.4135393
bifactor.b1_gender.male	b1cov.b1_age	
2.4152942	0.4394015	

Individual P-values (lower = worse):

edges	b1degree1	cycle4
0.59860632	0.33505316	0.67921157
bifactor.b1_gender.male	b1cov.b1_age	
0.01572251	0.66037066	

Joint P-value (lower = worse): 0.379943

Chain 8

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-2.865211	1.354826	-3.222967
bifactor.b1_gender.male	-1.976794	b1cov.b1_age	-3.141841

Individual P-values (lower = worse):

	edges	b1degree1	cycle4
	0.004167309	0.175472883	0.001268703
bifactor.b1_gender.male	0.048064957	b1cov.b1_age	0.001678894

Joint P-value (lower = worse): 0.04766405

Chain 9

Fraction in 1st window = 0.1

Fraction in 2nd window = 0.5

	edges	b1degree1	cycle4
	-0.9140193	0.1936344	-0.2219224
bifactor.b1_gender.male	-1.5686701	b1cov.b1_age	-0.7949470

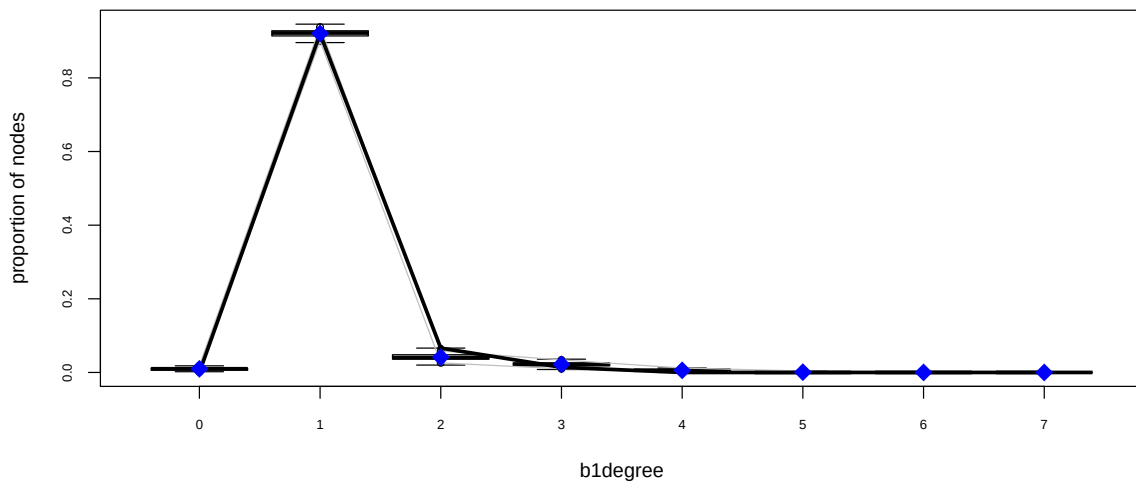
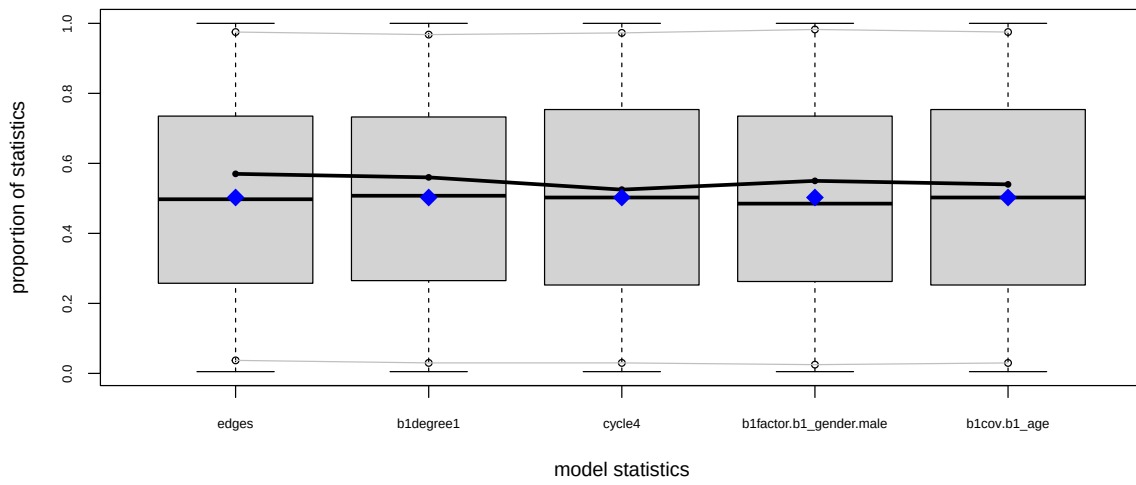
Individual P-values (lower = worse):

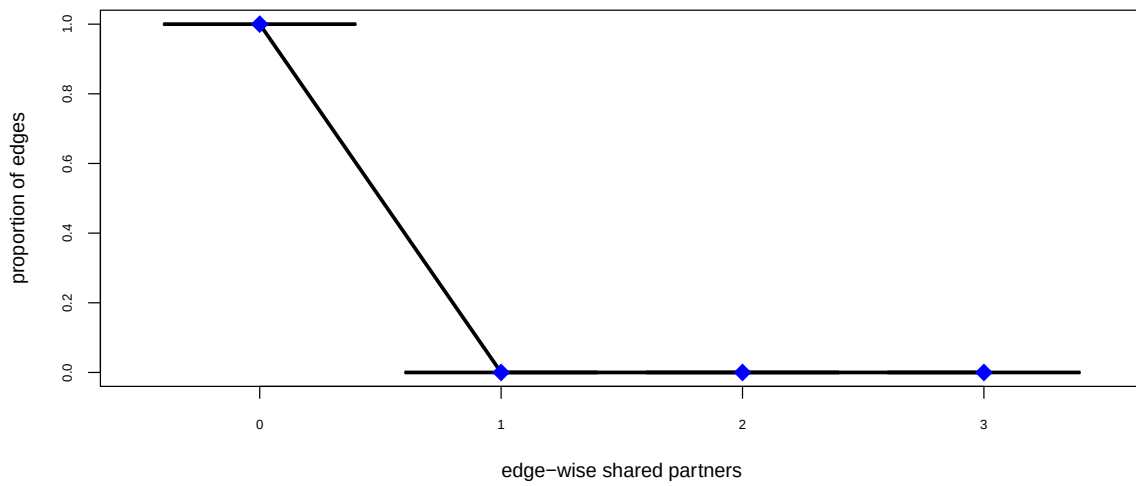
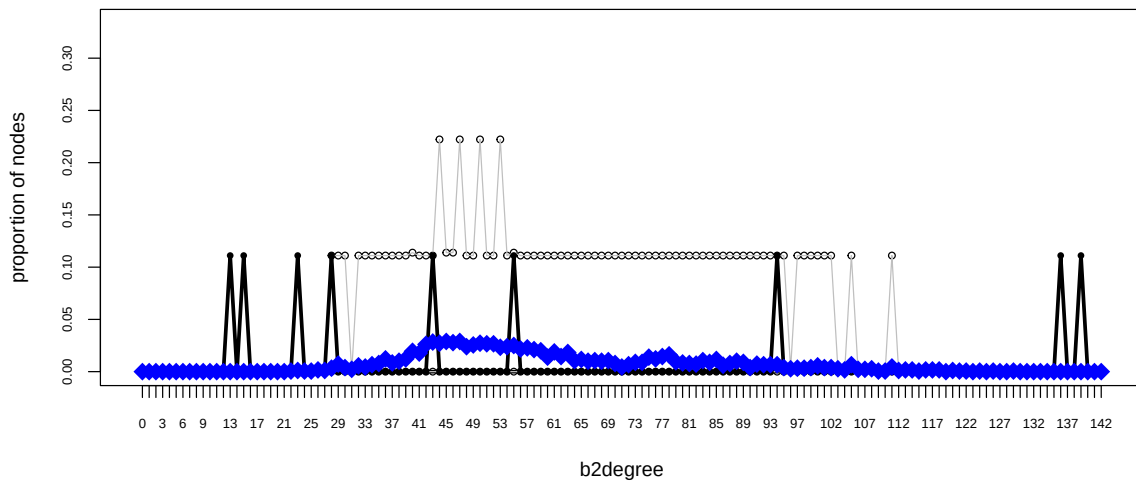
	edges	b1degree1	cycle4
	0.3607067	0.8464621	0.8243743
bifactor.b1_gender.male	0.1167248	b1cov.b1_age	0.4266443

Joint P-value (lower = worse): 0.2069806

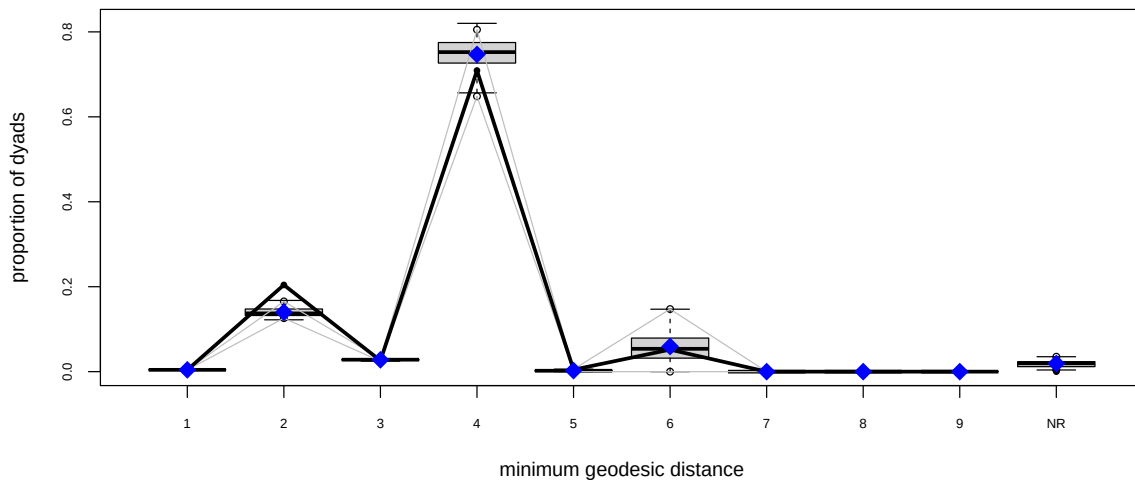
Note: To save space, only one in every 4 iterations of the MCMC sample used for estimation was stored for diagnostics. Sample size per chain was originally around 14928 with thinning interval 31.

Note: MCMC diagnostics shown here are from the last round of simulation, prior to computation of final parameter estimates. Because the final estimates are refinements of those used for this simulation run, these diagnostics may understate model performance. To directly assess the performance of the final model on in-model statistics, please use the GOF command: `gof(ergmFitObject, GOF=~model)`.





Goodness-of-fit diagnostics



Goodness-of-fit for model statistics

	obs	min	mean	max MC	p-value
edges	546	528	546.755	567	0.93
b1degree1	459	445	459.310	472	0.95
cycle4	67	29	68.370	148	0.98
b1factor.b1_gender.male	257	244	257.430	280	1.00
b1cov.b1_age	20959	20177	20992.900	21751	0.92

Goodness-of-fit for minimum geodesic distance

	obs	min	mean	max MC	p-value
1	546	528	546.755	567	0.93
2	26301	15719	18052.580	22225	0.00
3	3351	2939	3586.890	3925	0.27
4	91309	78517	96179.115	105613	0.32
5	594	0	313.345	948	0.16
6	6677	0	7631.630	24797	0.96
Inf	0	507	2467.685	6513	0.00

Goodness-of-fit for bipartition 1 degree

	obs	min	mean	max MC	p-value
0	0	1	4.890	13	0.00
1	459	445	459.310	472	0.95
2	33	10	20.650	33	0.01
3	7	3	10.900	18	0.22
4	0	0	2.840	9	0.11

5	0	0	0.380	3	1.00
6	0	0	0.025	1	1.00
7	0	0	0.005	1	1.00

Goodness-of-fit for bipartition 2 degree

	obs	min	mean	max	MC	p-value
13	1	0	0.000	0		0.00
15	1	0	0.000	0		0.00
22	0	0	0.005	1		1.00
23	1	0	0.010	1		0.02
24	0	0	0.005	1		1.00
25	0	0	0.005	1		1.00
26	0	0	0.015	1		1.00
27	0	0	0.010	1		1.00
28	1	0	0.030	1		0.06
29	0	0	0.065	2		1.00
30	0	0	0.035	2		1.00
31	0	0	0.020	1		1.00
32	0	0	0.050	1		1.00
33	0	0	0.040	1		1.00
34	0	0	0.060	2		1.00
35	0	0	0.070	2		1.00
36	0	0	0.110	2		1.00
37	0	0	0.075	2		1.00
38	0	0	0.090	1		1.00
39	0	0	0.110	2		1.00
40	0	0	0.175	2		1.00
41	0	0	0.160	2		1.00
42	0	0	0.235	2		1.00
43	1	0	0.255	2		0.48
44	0	0	0.245	2		1.00
45	0	0	0.260	2		1.00
46	0	0	0.250	2		1.00
47	0	0	0.260	2		1.00
48	0	0	0.215	2		1.00
49	0	0	0.230	2		1.00
50	0	0	0.245	2		1.00
51	0	0	0.240	2		1.00
52	0	0	0.240	2		1.00
53	0	0	0.210	3		1.00
54	0	0	0.215	2		1.00
55	1	0	0.225	2		0.40
56	0	0	0.195	2		1.00
57	0	0	0.205	2		1.00
58	0	0	0.190	2		1.00

59	0	0	0.180	2	1.00
60	0	0	0.125	2	1.00
61	0	0	0.170	2	1.00
62	0	0	0.130	2	1.00
63	0	0	0.165	2	1.00
64	0	0	0.095	1	1.00
65	0	0	0.105	3	1.00
66	0	0	0.090	1	1.00
67	0	0	0.095	2	1.00
68	0	0	0.090	2	1.00
69	0	0	0.095	2	1.00
70	0	0	0.070	1	1.00
71	0	0	0.040	1	1.00
72	0	0	0.060	1	1.00
73	0	0	0.080	2	1.00
74	0	0	0.080	1	1.00
75	0	0	0.125	2	1.00
76	0	0	0.110	2	1.00
77	0	0	0.130	2	1.00
78	0	0	0.145	1	1.00
79	0	0	0.085	1	1.00
80	0	0	0.075	1	1.00
81	0	0	0.070	1	1.00
82	0	0	0.065	1	1.00
83	0	0	0.095	1	1.00
84	0	0	0.080	2	1.00
85	0	0	0.105	2	1.00
86	0	0	0.055	1	1.00
87	0	0	0.070	1	1.00
88	0	0	0.090	1	1.00
89	0	0	0.080	1	1.00
90	0	0	0.035	1	1.00
91	0	0	0.065	1	1.00
92	0	0	0.060	1	1.00
93	0	0	0.060	1	1.00
94	1	0	0.060	1	0.12
95	0	0	0.035	2	1.00
96	0	0	0.025	1	1.00
97	0	0	0.030	1	1.00
98	0	0	0.030	1	1.00
99	0	0	0.035	1	1.00
100	0	0	0.050	2	1.00
101	0	0	0.035	1	1.00
102	0	0	0.035	1	1.00
103	0	0	0.025	1	1.00
104	0	0	0.015	1	1.00

105	0	0	0.060	2	1.00
106	0	0	0.020	1	1.00
107	0	0	0.020	1	1.00
108	0	0	0.025	1	1.00
109	0	0	0.005	1	1.00
110	0	0	0.005	1	1.00
111	0	0	0.040	2	1.00
112	0	0	0.010	1	1.00
113	0	0	0.015	1	1.00
114	0	0	0.015	1	1.00
115	0	0	0.005	1	1.00
116	0	0	0.015	1	1.00
117	0	0	0.015	1	1.00
118	0	0	0.015	1	1.00
120	0	0	0.010	1	1.00
121	0	0	0.005	1	1.00
122	0	0	0.005	1	1.00
124	0	0	0.005	1	1.00
126	0	0	0.005	1	1.00
129	0	0	0.005	1	1.00
136	1	0	0.000	0	0.00
139	1	0	0.000	0	0.00

Goodness-of-fit for edgewise shared partner

obs	min	mean	max MC	p-value
546.000	528.000	546.755	567.000	0.930

A.5 Source Code - Data Preprocessing

```

1 # ----- #
2 install.packages("viridis") # For Colours
3 install.packages("here") # To locate files from RProj
4 set.seed(42)
5 # ----- #
6 # Import the Bondora P2P Dataset
7 obj_paths = "resources/objects/"
8 bondora_raw <- read.csv("dataset/LoanData_Bondora.csv")
9 raw_cols <- colnames(bondora_raw)
10 # ----- #
11 # Select Columns to Keep
12 keep_cols <- c("LoanId", "UserName", "Age", "Gender",
13               "Country", "Amount", "Interest", "LoanDuration",
14               "UseOfLoan", "Rating", "Restructured", "MonthlyPayment",
15               "OccupationArea", "BiddingStartedOn")
16
17 bondora <- bondora_raw[keep_cols]
18
19 # Change date format to the correct one
20 bondora$BiddingStartedOn <- as.POSIXct(bondora$BiddingStartedOn,
21                                       format = "%Y-%m-%d %H:%M:%S")
22 bondora$year <- as.numeric(format(bondora$BiddingStartedOn, "%Y"))
23
24 # Remove Rows with any NAs -> Complete Dataset Preferred
25 print(paste("NA Count |", sum(is.na(bondora)), "rows"))
26 bondora <- na.omit(bondora)
27 print(paste("NA Count |", sum(is.na(bondora)), "rows"))
28
29 # Observe the distribution of Loan Use over Time
30 unique_counts <- tapply(bondora$UserName,
31                          list(bondora$year, bondora$UseOfLoan),
32                          function(x) length(unique(x)))
33 unique_counts[is.na(unique_counts)] <- 0
34
35 barplot(t(unique_counts), # transpose so bars = loan types
36         beside = TRUE, # side-by-side bars
37         col = viridis::viridis(ncol(unique_counts)),
38         legend.text = colnames(unique_counts),
39         args.legend = list(x = "topright", cex = 0.8),
40         xlab = "Year",
41         ylab = "Number of Unique Users")
42 user_counts_plt <- recordPlot()
43 saveRDS(user_counts_plt,
44         here::here("resources", "objects", "preprocessing", "user_cnt_plt.Rds"))
45
46 # Restrict to Recent Time Period
47 bondora_test <- bondora[bondora$year > 2013 & bondora$year < 2017, ]
48
49 # See the distribution of LoanUses
50 see_counts <- function(var1, var2) {
51   counts <- table(var1, var2)
52   total_counts <- colSums(counts)
53   prop <- total_counts / sum(total_counts)
54
55   return(list(total_counts, round(prop, 2)))
56 }
57 before_counts <- see_counts(bondora_test$UserName, bondora_test$UseOfLoan)
58 barplot(before_counts[[1]])
59
60 # Number of unique users in the subsample
61 unique_users <- unique(bondora_test$UserName)
62 sample_users <- sample(unique_users, 500)
63 bondora_clean <- bondora_test[bondora_test$UserName %in% sample_users, ]
64
65 after_counts <- see_counts(bondora_clean$UserName, bondora_clean$UseOfLoan)

```

```

66 | barplot(after_counts[[1]])
67 |
68 | # DEPRECATED METHOD
69 | # Filter People with 2+ Loans
70 | #user_counts <- table(bondora_test$UserName)
71 | #multi_users <- names(user_counts[user_counts > 2])
72 | #bondora_clean <- bondora_test[bondora_test$UserName %in% multi_users, ]
73 |
74 | # DEPRECATED METHOD
75 | # Remove Users with only One Loan
76 | #user_counts <- table(bondora_clean$UserName)
77 | #multi_users <- names(user_counts[user_counts > 5])
78 | #bondora_clean <- bondora_clean[bondora_clean$UserName %in% multi_users, ]
79 | #bondora_test <- bondora_raw[bondora_raw$UserName %in% multi_users, ]
80 |
81 | # See if Ratings are Properly Encoded
82 | unique(bondora_clean$Rating)
83 | bondora_clean <- bondora_clean[bondora_clean$Rating != "", ]
84 |
85 | # Extract UseofLoan Types and Turn into Factor
86 | bondora_clean$UseOfLoan_factor <- as.factor(bondora_clean$UseOfLoan)
87 | unique(bondora_clean$UseOfLoan_factor)
88 |
89 | loan_use_labels <- c(
90 |   "-1" = "NA",
91 |   "0" = "Loan Consolidation",
92 |   "1" = "Real Estate",
93 |   "2" = "Home Improvement",
94 |   "3" = "Business",
95 |   "4" = "Education",
96 |   "5" = "Travel",
97 |   "6" = "Vehicle",
98 |   "7" = "Other",
99 |   "8" = "Health",
100 |   "110" = "Other Business",
101 |   "102" = "Undefined Business",
102 |   "108" = "Undefined Business"
103 | )
104 | # Change Labels for Cleaned Dataset
105 | bondora_clean$UseOfLoan_factor <- loan_use_labels[
106 |   as.character(bondora_clean$UseOfLoan)]
107 |
108 | # Change Labels for Uncleaned Dataset
109 | bondora_test$UseOfLoan_factor <- loan_use_labels[
110 |   as.character(bondora_test$UseOfLoan)]
111 |
112 | # Add labels to the OccupationArea Variable
113 | levels(as.factor(bondora_clean$OccupationArea)) # view codes
114 |
115 | occupation_labels <- c(
116 |   "-1" = "NA",
117 |   "1" = "Other",
118 |   "2" = "Mining",
119 |   "3" = "Processing",
120 |   "4" = "Energy",
121 |   "5" = "Utilities",
122 |   "6" = "Construction",
123 |   "7" = "Retail",
124 |   "8" = "Transport",
125 |   "9" = "Hospitality",
126 |   "10" = "Info/Telcom",
127 |   "11" = "Finance",
128 |   "12" = "Realestate",
129 |   "13" = "Research",
130 |   "14" = "Admin",
131 |   "15" = "Civil/Mil",

```

```

132 "16" = "Education",
133 "17" = "Healthcare",
134 "18" = "Social",
135 "19" = "Art/Ent",
136 "20" = "Agriculture",
137 "21" = "Forestry/Fish"
138 )
139 # store original
140 bondora_clean$occupation_code <- bondora_clean$OccupationArea
141 bondora_clean$occupation_label <- occupation_labels[
142   as.character(bondora_clean$OccupationArea)]
143
144 bondora_test$occupation_code <- bondora_test$OccupationArea
145 bondora_test$occupation_label <- occupation_labels[
146   as.character(bondora_test$OccupationArea)]
147 # ----- #
148 # Observe Descriptive Statistics
149
150 cols <- viridis::viridis(30)
151
152 # Make function to save plots
153 save_plot <- function(plt_nam) {
154   plt <- recordPlot()
155   saveRDS(plt, here::here("resources", "objects", "preprocessing",
156     paste0(plt_nam, ".Rds")))
157 }
158
159 # Make function to consistently plot comparisons
160 plot_desc_hists <- function(df1, df2, col_name, type) {
161
162   par(mfrow=c(1,3))
163
164   hist(df1[[col_name]], xlab=type, col=cols[15], main="", breaks=10)
165   mtext("Initial Sample", side=3, adj=0, line=0.25, cex=1, font=2)
166
167   hist(df2[[col_name]], xlab=type, col=cols[15], main="", breaks=10)
168   mtext("Reduced Sample", side=3, adj=0, line=0.25, cex=1, font=2)
169
170   qqplot(df1[[col_name]], df2[[col_name]], main="", cex=1,
171     xlab="Full Sample", ylab="Subsample", line=0.25)
172   abline(0, 1, lty=2)
173
174   mtext("QQ Plot", side=3, adj=0, line=0.25, cex=1, font=2)
175
176   mtext(type, outer = TRUE, line = -2, side=3, cex = 1.3, font = 2)
177
178   # Reset plot window
179   par(mfrow=c(1,1), mar=c(5,4,4,2)+0.1)
180 }
181
182 plot_desc_bar <- function(df1, df2, col_name, type) {
183
184   par(mfrow=c(1,2), mar=c(5,10,4,2))
185
186   barplot(sort(table(df1[[col_name]]), decreasing = F),
187     xlab=type, col=cols[15], horiz=TRUE, las=1)
188   mtext("Initial Sample", side=3, adj=0, line=0.25, cex=1, font=2)
189
190   barplot(sort(table(df2[[col_name]]), decreasing = F),
191     xlab=type, col=cols[15], horiz=TRUE, las=1)
192   mtext("Reduced Sample", side=3, adj=0, line=0.25, cex=1, font=2)
193
194   mtext(type, outer = TRUE, line = -2, side=3, cex = 1.3, font = 2)
195
196   # Reset plot window
197   par(mfrow=c(1,1), mar=c(5,4,4,2)+0.1)

```



```

198 }
199
200 plot_desc_hists(bondora_test, bondora_clean, "Amount", "Amount")
201 save_plot("hist_amt")
202
203 plot_desc_hists(bondora_test, bondora_clean, "Interest", "Interest")
204 save_plot("hist_int")
205
206 plot_desc_hists(bondora_test, bondora_clean, "LoanDuration", "Loan Duration")
207 save_plot("hist_loandur")
208
209 plot_desc_hists(bondora_test, bondora_clean, "MonthlyPayment", "Monthly Payment")
210 save_plot("hist_monpmt")
211
212 plot_desc_hists(bondora_test, bondora_clean, "Age", "Age")
213 save_plot("hist_age")
214
215 plot_desc_bar(bondora_test, bondora_clean, "UseOfLoan_factor", "Loan Purpose")
216 save_plot("bar_loanuse")
217
218 plot_desc_bar(bondora_test, bondora_clean, "Rating", "Credit Rating")
219 save_plot("bar_rating")
220
221 plot_desc_bar(bondora_test, bondora_clean, "occupation_label", "Occupation")
222 save_plot("bar_occupation")
223
224 # Get Tabular Summary Statistics
225 tab_comps <- function(df1, df2, cols) {
226   stats <- c("Mean"=mean, "Median"=median, "Std. Dev."=sd, "Min"=min, "Max"=max)
227
228   get_stats <- function(d) {
229     t(sapply(d[cols], function(x)
230       sapply(stats, function(f) round(f(x, na.rm=TRUE), 2))
231     ))
232   }
233
234   df1_stats <- get_stats(df1)
235   df2_stats <- get_stats(df2)
236
237   out <- rbind(
238     cbind(DataFrame = "Initial Sample",
239           Variable = rownames(df1_stats), df1_stats),
240     cbind(DataFrame = "Reduced Sample",
241           Variable = rownames(df2_stats), df2_stats)
242   )
243   rownames(out) <- NULL
244   as.data.frame(out)
245 }
246
247 tab_results <- tab_comps(bondora_test, bondora_clean,
248   c("Amount", "Interest", "Age", "LoanDuration"))
249
250 knitr::kable(tab_results)
251
252 # Save table for use in the Report
253 saveRDS(tab_results,
254   file=paste0(obj_paths, "preprocessing/", "summary_table.Rds"))
255 # ----- #
256 # Convert Dataset into Incidence Matrix to form Network Object (for ERGM)
257 bondora_slim <- bondora_clean
258
259 # Create the Incidence Matrix for Use of Loan
260 bondora_matrix <- table(
261   bondora_slim$UserName, bondora_slim$UseOfLoan)
262 bondora_matrix[bondora_matrix > 0] <- 1 # Given that ergm.counts fails with GOF
263

```

```

264 # Create network object with counts as Edge attribute
265 bondora_net <- network::network(
266   bondora_matrix, directed=FALSE, bipartite=nrow(bondora_matrix),
267   ignore.eval = FALSE, names.eval="frequency", loops=FALSE)
268
269 # Set the bipartite Attribute - UNNECESSARY GIVEN bipartite=length(n)
270 len <- dim(bondora_matrix)[1]
271 len_b2 <- dim(bondora_matrix)[2]
272 b_indicator <- c(rep(1,len),rep(2,len_b2))
273 network::set.vertex.attribute(bondora_net,
274                               "bipartite",value=b_indicator)
275
276 # Extract Partition 2 Labels
277 loan_use <- levels(bondora_clean$UseOfLoan_factor)
278
279 # Create Loan Type Attribute for Partition 2
280 b2_loantype <- rep(NA, len)
281 b2_loantype <- c(b2_loantype, loan_use)
282
283 if (length(b2_loantype) == network::network.size(bondora_net)) {
284   network::set.vertex.attribute(
285     bondora_net, "b2_loantype", value = b2_loantype)
286 }
287
288 # Add Age Vertex Attribute to B1
289 age <- bondora_clean$Age[match(
290   rownames(bondora_matrix), bondora_clean$UserName)]
291 b1_age <- c(age, rep(NA, len_b2))
292 network::set.vertex.attribute(
293   bondora_net, "b1_age", value=b1_age
294 )
295
296 # Add Gender Vertex Attribute to B1
297 gender <- bondora_clean$Gender[match(
298   rownames(bondora_matrix), bondora_clean$UserName)]
299 unique(gender) # Check to see if encoded properly
300 gender <- ifelse(gender == 0, "male", "female")
301 b1_gender <- c(gender, rep(NA, len_b2))
302 network::set.vertex.attribute(
303   bondora_net, "b1_gender", value = b1_gender
304 )
305
306 # Save the network and data frame object
307 saveRDS(bondora_slim, file=paste0(obj_paths, "preprocessing/", "bondora_df.Rds"))
308 saveRDS(bondora_net, file=paste0(obj_paths, "preprocessing/", "bondora_net.Rds"))
309 # ----- #
310 # First, create Incidence Matrix again but with Frequency Counts
311 loan_use_matrix <- table(
312   bondora_slim$UserName, bondora_slim$UseOfLoan)
313
314 # Get the Adjacency Matrix for Loan Use Similarity (Dependent QAP Variable)
315 adj_mat_loan_use <- loan_use_matrix %*% t(loan_use_matrix)
316 # Remove self weights to remove any loops
317 diag(adj_mat_loan_use) <- 0
318
319 # Get the Adjacency Matrix for Credit Rating Similarity (Predictor in QAP)
320 incidence_rating <- table(bondora_slim$UserName, bondora_slim$Rating)
321 adj_mat_rating <- incidence_rating %*% t(incidence_rating)
322 diag(adj_mat_rating) <- 0
323
324 # Get Adjacency Matrix for Occupation Similarity (Predictor in QAP, Binary)
325 incidence_occupation <- table(bondora_slim$UserName,
326                               bondora_slim$occupation_label)
327 adj_mat_occupation <- incidence_occupation %*% t(incidence_occupation)
328 adj_mat_occupation[adj_mat_occupation > 0] <- 1
329 diag(adj_mat_occupation) <- 0

```

```

330
331 # Get the Adjacency Matrix for Loan Amount (Control in QAP) - BINARY
332 # First, bin the Loan Amounts
333 bondora_slim$Amount_bins <- cut(
334   bondora_slim$Amount, breaks=c(0,2000,4000,6000,8000,10000),
335   labels = c(1:5)
336 )
337 incidence_amount_bins <- table(bondora_slim$UserName, bondora_slim$Amount_bins)
338 adj_mat_amount_bins <- incidence_amount_bins %>% t(incidence_amount_bins)
339 diag(adj_mat_amount_bins) <- 0
340
341 # Continous Absolute Difference Approach
342 avg_amount <- tapply(bondora_slim$Amount, bondora_slim$UserName, mean)
343 adj_mat_amount_diff <- outer(avg_amount, avg_amount,
344   FUN = function(x,y) abs(x - y))
345 diag(adj_mat_amount_diff) <- 0
346
347 # Get Matrix for Differences in Age
348 incidence_age <- table(bondora_slim$UserName, bondora_slim$Age)
349 borrower_ages <- as.numeric(colnames(incidence_age)[max.col(incidence_age)])
350 names(borrower_ages) <- rownames(incidence_age)
351 adj_mat_age <- outer(borrower_ages, borrower_ages,
352   FUN = function(x, y) abs(x - y))
353 rownames(adj_mat_age) <- colnames(adj_mat_age) <- names(borrower_ages)
354
355 # Get Adjacency Matrix for (same) Gender
356 incidence_gender <- table(bondora_slim$UserName, bondora_slim$Gender)
357 adj_mat_gender <- incidence_gender %>% t(incidence_gender)
358 # Make the matrix binary for homophily
359 adj_mat_gender <- ifelse(adj_mat_gender > 0, 1, 0)
360 diag(adj_mat_gender) <- 0
361
362 # Get Adjacency Matrix for Differences in Average Loan Duration
363 borrower_loandur <- sapply(tapply(bondora_slim$LoanDuration,
364   bondora_slim$UserName, unique),
365   mean)
366 adj_mat_loandur_diff <- outer(borrower_loandur, borrower_loandur,
367   FUN=function(x,y) abs(x-y))
368 diag(adj_mat_loandur_diff) <- 0
369
370 # Get Adjacency Matrix for Homophily in Restructure of Loans
371 incidence_restructure <- table(bondora_slim$UserName, bondora_slim$Restructured)
372 adj_mat_rest <- incidence_restructure %>% t(incidence_restructure)
373 diag(adj_mat_rest) <- 0
374
375 # Save the objects for the QAP Regression in different Script
376 qap_paths = paste0(obj_paths, "/qap/")
377 saveRDS(b_indicator, file=paste0(
378   "resources/objects/preprocessing/indicator.Rds"))
379
380 saveRDS(adj_mat_loan_use, file=paste0(qap_paths, "adj_mat_loanuse.Rds"))
381 saveRDS(adj_mat_occupation, file=paste0(qap_paths, "adj_mat_occup.Rds"))
382 saveRDS(adj_mat_rating, file=paste0(qap_paths, "adj_mat_rating.Rds"))
383 saveRDS(adj_mat_amount_diff, file=paste0(qap_paths, "adj_mat_amtdiffs.Rds"))
384 saveRDS(adj_mat_age, file=paste0(qap_paths, "adj_mat_agediffs.Rds"))
385 saveRDS(adj_mat_gender, file=paste0(qap_paths, "adj_mat_gender.Rds"))
386 saveRDS(adj_mat_loandur_diff, file=paste0(qap_paths, "adj_mat_loandurdiffs.Rds"))
387 saveRDS(adj_mat_rest, file=paste0(qap_paths, "adj_mat_rest.Rds"))
388 # ----- #

```

scripts/bondora_preprocessing.R

A.6 Source Code - QAP Linear Regression

```
1 # ----- #
2 # NOTE: NOT THE LATEST QAP ANALYSIS!
3
4 # If not already Installed
5 install.packages("viridis") # For Colours
6 install.packages("here") # To locate files from RProj
7 install.packages("gridExtra") # For multiple ggplot2 plots
8
9 # Set colour palette
10 cols <- viridis::viridis(30)
11
12 # Ensure repeatability
13 set.seed(42)
14 # ----- #
15 # Load Relevant Files
16 qap_path="resources/objects/qap/"
17
18 loan_use_mat <- readRDS(paste0(qap_path, "adj_mat_loanuse.RDS"))
19 rating_mat <- readRDS(paste0(qap_path, "adj_mat_rating.RDS"))
20 amt_diffs_mat <- readRDS(paste0(qap_path, "adj_mat_amtdiffs.RDS"))
21 age_diffs_mat <- readRDS(paste0(qap_path, "adj_mat_agediffs.RDS"))
22 gender_mat <- readRDS(paste0(qap_path, "adj_mat_gender.RDS"))
23 loandur_diffs_mat <- readRDS(paste0(qap_path, "adj_mat_loandurdiffs.RDS"))
24 rest_mat <- readRDS(paste0(qap_path, "adj_mat_rest.RDS"))
25 occup_mat <- readRDS(paste0(qap_path, "adj_mat_occup.Rds"))
26 # ----- #
27 # Create function to determine significance from t-value statistic
28 t_to_stars <- function(t) {
29   stars <- rep("", length(t))
30   stars[abs(t) >= 1.96] <- "*"
31   stars[abs(t) >= 2.576] <- "**"
32   stars[abs(t) >= 3.291] <- "***"
33
34   return(stars)
35 }
36
37 # Make function to save plots
38 save_qap_plot <- function(plt_nam) {
39   plt <- recordPlot()
40   saveRDS(plt, here::here("resources", "objects", "qap",
41     paste0(plt_nam, ".Rds")))
42 }
43
44 # Make function to run QAPs automatically
45 run_QAP <- function(y_mat, xlist, varnames, model_name) {
46
47   # Run QAP Linear Regression
48   model <- sna::netlm(y = y_mat, x = xlist, nullhyp = "qapspp", reps = 3000)
49   model$names <- varnames
50   summary(model)
51
52   # Obtain results from Model and Name them
53   results <- model$coefficients
54   names(results) <- varnames
55
56   # Add significance stars to model results
57   results_sig <- paste(round(results, 3), t_to_stars(model$tstat))
58
59   # Save Model
60   saveRDS(model, here::here("resources", "objects", "qap",
61     paste0(model_name, ".Rds")))
62
63   # Prepare function output
64   list_names <- c("model", "results", "results_sig")
65   items <- list(model, results, results_sig)
```

```

66   names(items) <- list_names
67
68   return(items)
69 }
70
71 # Make function to Plot QAP Coefficients
72 plot_coef_qap <- function(model,
73                             model_name = "Model Estimates",
74                             sub = "Subtitle",
75                             x_axis_label = "Coefficient Estimate") {
76
77   # Extract Coefficients and SEs from ERM
78   coefs <- model$coefficients
79   ses <- model$coefficients / model$tstat
80
81   # Make DF
82   df <- data.frame(
83     term = model$names,
84     estimate = coefs,
85     se = ses
86   )
87
88   df$lower <- df$estimate - 1.96 * df$se
89   df$upper <- df$estimate + 1.96 * df$se
90
91   df$sig <- df$lower * df$upper > 0
92   df$sig <- as.character(df$sig) # Convert to character to make it work
93
94   order_index <- order(df$estimate)
95   df$term <- factor(df$term, levels = df$term[order_index])
96
97   ggplot2::ggplot(df, ggplot2::aes(x = estimate, y = term)) +
98     ggplot2::geom_vline(xintercept = 0, linetype = "dashed", color = "grey40") +
99     ggplot2::geom_errorbarh(ggplot2::aes(xmin = lower, xmax = upper),
100                           height = 0.2) +
101     ggplot2::geom_point(size = 3) +
102     ggplot2::theme_bw() +
103     ggplot2::theme(
104       panel.grid.major.x = ggplot2::element_blank(),
105       panel.grid.minor.x = ggplot2::element_blank()) +
106     ggplot2::labs(
107       x = x_axis_label,
108       y = "Variable",
109       title = model_name,
110       subtitle = sub
111       #color = "Significance"
112     )
113 }
114 # Make Function to Plot Diagnostics
115 # DEPRECATED SINCE PROGRESS MEETING 3
116 qap_diags <- function(model, title, filename) {
117   resid <- model$model$residuals
118   fitted <- model$model$fitted.values
119
120   # Set viewing window to two plots
121   par(mfrow=c(1,2))
122
123   # Residuals Plot
124   hist(resid, xlab="Residuals",
125        main=title)
126
127   # Fitted vs Residuals
128   plot(fitted, resid, xlab="Fitted Values", ylab="Residuals",
129        main=title)
130
131   # Capture and Save the Plot

```

```

132   save_qap_plot(filename)
133
134   # Reset plot view
135   par(mfrow=c(1,1))
136 }
137
138 var_names <- c("Intercept", "Rating", "Occupation", "Loan Amount", "Age",
139               "Gender", "Loan Duration", "Restructured")
140 main_pred_names <- c("Intercept", "Rating", "Occupation")
141 main_pred_vars <- list(rating_mat, occup_mat)
142 all_pred_vars <- list(rating_mat, occup_mat, amt_diffs_mat, age_diffs_mat,
143                      gender_mat, loandur_diffs_mat, rest_mat)
144 # ----- #
145 # Basic QAP Linear Regression 1 - Unstandardised + No Controls
146 qap_m1 <- run_QAP(loan_use_mat, main_pred_vars, main_pred_names, "qap_m1")
147 summary(qap_m1$model)
148
149 # Plot the QAP Model Coefficients and Save it
150 qap_m1_coef_plot <- plot_coef_qap(qap_m1$model,
151                                  model_name = "MRQAP Model 1",
152                                  sub = "No Controls")
153 saveRDS(qap_m1_coef_plot,
154         here::here("resources", "objects", "qap", "qap_m1_coef_plot.Rds"))
155
156 # Run Diagnostics Plots - DEPRECATED SINCE PROGRESS MEETING 3 SUGGESTIONS
157 #qap_diags(qap_m1, "QAP LR - Unstandardised + No Controls", "qap_m1_diags")
158
159 # ----- #
160 # Basic QAP Linear Regression 2 - Unstandardised + Controls
161 qap_m3 <- run_QAP(loan_use_mat, all_pred_vars, var_names, "qap_m3")
162
163 summary(qap_m3$model)
164
165 # Plot the QAP Model Coefficients and Save it
166 qap_m3_coef_plot <- plot_coef_qap(qap_m3$model,
167                                  model_name = "MRQAP Model 2",
168                                  sub = "With Controls")
169 saveRDS(qap_m3_coef_plot,
170         here::here("resources", "objects", "qap", "qap_m3_coef_plot.Rds"))
171
172 # Arrange the Plots together
173 qap_coef_plots <- gridExtra::grid.arrange(qap_m1_coef_plot, qap_m3_coef_plot,
174                                           ncol = 2)
175 # Save the Plots
176 saveRDS(qap_coef_plots,
177         here::here("resources", "objects", "qap", "qap_coef_plots.Rds"))
178
179 # Run Diagnostics Plots - DEPRECATED SINCE PROGRESS MEETING 3 SUGGESTIONS
180 #qap_diags(qap_m3, "QAP LR - Unstandardised + Controls", "qap_m3_diags")
181
182 # ----- #
183 # Collect Results in Table
184
185 # Extract R2 from the netlm object from the Console output
186 # It was too difficult to calculate R2 from scratch and netlm does not
187 # provide this as an output directly from the netlm object
188 calc_r2 <- function(model) {
189
190   out <- capture.output(model)
191   r2_line <- grep("Multiple R-squared", out, value = TRUE)
192   r2 <- as.numeric(
193     sub(".*Multiple R-squared:\\s*([0-9\\.]+).*", "\\1", r2_line))
194
195   return(r2)
196 }
197

```

```

198 # Make function to get details of netlm manually
199 extract.netlm <- function(model) {
200   tr <- texreg::createTexreg(
201     coef.names = model$names,
202     coef = model$coefficients,
203     se = model$coefficients / model$tstat,
204     pvalues = model$pgregabs,
205     gof.names = c("R-squared"),
206     gof = (calc_r2(model)),
207     gof.decimal = c(TRUE)
208   )
209   return(tr)
210 }
211
212 # View the Table of Results
213 titles <- c("M1 | No Controls", "M2 | Controls")
214 extracted_models <- lapply(list(qap_m1$model, qap_m3$model), extract.netlm)
215 texreg::screenreg(extracted_models, custom.model.names = titles, digits = 3)
216
217 # Save the table for use in the Report
218 saveRDS(list(extracted_models, titles),
219         here::here("resources", "objects", "qap", "qap_tables.Rds"))
220 # ----- #

```

scripts/qap_network_analysis.R

A.7 Source Code - ERGM Network Analysis

```
1 # ----- #
2 # NOTE: NOT THE LATEST ERGM ANALYSIS! FOR REFERENCE ONLY
3
4 # If not already Installed
5 install.packages("viridis") # For Colours
6 install.packages("Rglpk") # Additional solver for ERGMs
7 install.packages("here") # To locate files from RProj
8 install.packages("gridExtra") # To display multiple ggplots side by side
9
10 # Import the Network and Other Object
11 ergm_path <- "resources/objects/ergm/"
12 bondora_net <- readRDS(here::here(
13   "resources", "objects", "preprocessing", "bondora_net.Rds"))
14 b_indicator <- readRDS(here::here(
15   "resources", "objects", "preprocessing", "indicator.Rds"))
16
17 # Set colour palette
18 cols <- viridis::viridis(30)
19
20 # Determine acceptable core count
21 n_cores <- parallel::detectCores() - 3 # Leave some out for other processes
22 print(paste("You have", n_cores, "usable cores"))
23
24 # Repeatability
25 set.seed(42)
26
27 # Save plots
28 save_ergm_plot <- function(plt_nam) {
29   plt <- recordPlot()
30   saveRDS(plt, here::here("resources", "objects", "ergm",
31     paste0(plt_nam, ".Rds")))
32 }
33 # ----- #
34 # Copy network for plotting
35 bondora_plot <- bondora_net
36
37 # Get node type for plotting
38 type_indicator <- ifelse(b_indicator == 2, TRUE, FALSE)
39 shape <- ifelse(type_indicator, "square", "circle")
40 network::set.vertex.attribute(bondora_plot, "shape", shape)
41
42 # Get Category Count for Vertex Size
43 counts <- sna::degree(bondora_plot)
44 counts_att <- ifelse(type_indicator, log(counts)*4, counts*2.5)
45 network::set.vertex.attribute(bondora_plot, "size", counts_att)
46
47 # Colours for the Node Types
48 plot_cols <- ifelse(type_indicator, cols[5], cols[30])
49 network::set.vertex.attribute(bondora_plot, "color", plot_cols)
50
51 # Legend Plotting
52 type_legend <- ifelse(type_indicator, "Borrowers", "Loan Type")
53 type_legend <- as.factor(type_legend)
54
55 # Plot the Network
56 plot(snafun::to_igraph(bondora_plot),
57   #main = "Bipartite User-LoanUse",
58   edge.arrow.size = 0.3,
59   edge.color = rgb(0,0,0, alpha = 0.35),
60   vertex.frame.color = "black",
61   vertex.label = NA,
62   vertex.frame.size = 3,
63   edge.curved = FALSE,
64   layout=igraph::layout.fruchterman.reingold)
65 legend("bottomleft",
```



```

66     legend = levels(type_legend),
67     inset = c(0.15, 0.01),
68     col = c(cols[30], cols[5]),
69     pch = c(16, 15),
70     title = "Node Partitions",
71     title.font = 2,
72     cex = 1,          # Increase the text size
73     pt.cex = 2,       # Increase the point symbol size
74     box.lwd = 1,      # Thin box border
75     box.col = "black", # Box color
76     bty = "o"         # Use a box around legend
77 )
78 save_ergm_plot("network_plot")
79
80 # Summary Statistics and Save Them
81 density <- snafun::g_density(bondora_net)[1]
82 centralization <- snafun::g_centralize(bondora_net)[1]
83 vertices <- snafun::count_vertices(bondora_net)[1]
84 edges <- snafun::count_edges(bondora_net)[1]
85 dist <- snafun::g_mean_distance(bondora_net)[1]
86
87 net_names <- c("Vertex Count", "Edge Count", "Density", "Centralization",
88               "Mean Distance")
89 net_stats <- c(vertices, edges, density, centralization, dist)
90 net_stats <- sapply(net_stats, function(x) round(as.numeric(x), 2))
91
92 net_summary <- data.frame(Statistic = net_names,
93                           "Measure" = net_stats)
94 knitr::kable(net_summary)
95 saveRDS(net_summary, here::here("resources", "objects", "ergm", "net_summary.Rds"))
96
97 # Plot Network Summary Statistics
98 snafun::plot_centralities(bondora_net)
99 save_ergm_plot("network_plots")
100 # ----- #
101 # Make function to calculate probabilities from log odds
102 l odds_to_prob <- function(l_odd) {
103   return(exp(l_odd) / (1 + exp(l_odd)))
104 }
105 # Make function to save ERGM object
106 save_ergm <- function(object, id) {
107   saveRDS(object, file=here::here(
108     "resources", "objects", "ergm", paste0(id, ".Rds")))
109 }
110 # Make function to conduct ERGMs automatically
111 auto_ergm <- function(model, mcmc, name) {
112
113   # Conducts the GOF Diagnostics and then saves the model,
114   # mcmc diagnostics and gof object in a list.
115   # This list can be imported as an .RDS object into the R environment
116
117   # Diagnostics
118   if (mcmc) {
119     ergm::mcmc.diagnostics(model)
120   }
121
122   # The GOF must be adjusted otherwise it takes too long
123   # We do not limit the GOF by changing its range of parameters
124   gof <- ergm::gof(model,
125                     control = ergm::control.gof.ergm(
126                       nsim = 200,
127                       MCMC.burnin = 5000,
128                       MCMC.interval = 1000,
129                       parallel = n_cores,
130                       parallel.type = "PSOCK"
131                     ))

```

```

132
133 # Return List to view each item separately
134 result <- list(model, gof)
135 names(result) <- c("model", "gof")
136 save_ergm(result, paste0(name, "_panel"))
137
138 return(result)
139 }
140 # Make Function to do Coefficient Plot
141 plot_coef <- function(model,
142                       model_name = "Model Estimates",
143                       sub = "Subtitle",
144                       x_axis_label = "Coefficient Estimate") {
145
146   # Extract Coefficients and SEs from ERGM
147   coefs <- coef(model)
148   ses <- sqrt(diag(vcov(model)))
149
150   # Make DF
151   df <- data.frame(
152     term = names(coefs),
153     estimate = coefs,
154     se = ses
155   )
156
157   df$lower <- df$estimate - 1.96 * df$se
158   df$upper <- df$estimate + 1.96 * df$se
159
160   df$sig <- df$lower * df$upper > 0
161   df$sig <- as.character(df$sig) # Convert to character to make it work
162
163   order_index <- order(df$estimate)
164   df$term <- factor(df$term, levels = df$term[order_index])
165
166   ggplot2::ggplot(df, ggplot2::aes(x = estimate, y = term)) +
167     ggplot2::geom_vline(xintercept = 0, linetype = "dashed", color = "grey40") +
168     ggplot2::geom_errorbarh(ggplot2::aes(xmin = lower, xmax = upper),
169                           height = 0.2) +
170     ggplot2::geom_point(size = 3) +
171     ggplot2::theme_bw() +
172     ggplot2::theme(
173       panel.grid.major.x = ggplot2::element_blank(),
174       panel.grid.minor.x = ggplot2::element_blank() +
175     ggplot2::labs(
176       x = x_axis_label,
177       y = "Variable",
178       title = model_name,
179       subtitle = sub
180       #color = "Significance"
181     )
182   )
183 }
184 # ----- #
185 # Find max degree
186 (max_deg <- max(summary(bondora_net ~ b2factor("b2_loantype"))))
187 # ----- #
188 # Base Model + GOF
189 formula_base_model <- bondora_net ~ edges
190 base_ergm <- ergm::ergm(formula_base_model)
191
192 base_ergm_panel <- auto_ergm(base_ergm, mcmc = FALSE, name = "ergm_base")
193
194 snafun::stat_plot_gof(base_ergm_panel$gof)
195 models = list(base_ergm)
196 texreg::screenreg(models)
197 # ----- #
198 # Iteration 1 + MCMC Diagnostics + GOF

```

```

198 model_1_params <- bondora_net ~ edges +
199   # See if individuals tend to pick one loan type
200   b1degree(1)
201
202 model_1 <- ergm::ergm(
203   model_1_params,
204   #constraints = ~ bd(minout = 0, maxout = 80),
205
206   control = ergm::control.ergm(
207     # Greater burn-in for cleaner result
208     MCMC.burnin = 20000,
209     # Greater sample size for greater stability
210     MCMC.samplesize = 100000,
211     seed = 42,
212     MCMC.interval = 1000,
213     # Only needed for convergence pvals to improve
214     MCME.maxit = 45,
215     # Smaller steps for stability
216     MCME.steplength = 0.25,
217     parallel = n_cores,
218     parallel.type = "PSOCK"
219   )
220 )
221
222 model_1_panel <- auto_ergm(model=model_1, mcmc=TRUE, name="ergm_m1")
223 model_1_panel$gof
224
225 snafun::stat_plot_gof(model_1_panel$gof)
226 save_ergm_plot("m1_gof")
227
228 texreg::screenreg(list(base_ergm, model_1))
229
230 # Plot Coefficients
231 m1_coef_plot <- plot_coef(model_1, model_name = "Model 1",
232   sub = "b1degree")
233 saveRDS(m1_coef_plot,
234   here::here("resources", "objects", "ergm", "m1_coef_plot.Rds"))
235 # ----- #
236 # Iteration 2 + MCMC Diagnostics + GOF
237 model_2_params <- bondora_net ~ edges + b1degree(1) +
238   # See if there is clustering around pairs of loan types
239   cycle(4)
240
241 model_2 <- ergm::ergm(
242   model_2_params,
243   #constraints = ~ bd(minout = 0, maxout = 80),
244
245   control = ergm::control.ergm(
246     # Greater burn-in for cleaner result
247     MCMC.burnin = 20000,
248     # Greater sample size for greater stability
249     MCMC.samplesize = 100000,
250     seed = 42,
251     MCMC.interval = 1000,
252     # Only needed for convergence pvals to improve
253     MCME.maxit = 45,
254     # Smaller steps for stability
255     MCME.steplength = 0.25,
256     parallel = n_cores,
257     parallel.type = "PSOCK"
258   )
259 )
260
261 model_2_panel <- auto_ergm(model=model_2, mcmc=TRUE, name="ergm_m2")
262 snafun::stat_plot_gof(model_2_panel$gof)
263 model_2_panel$gof

```

```

264 models <- list(base_ergm, model_1, model_2)
265 texreg::screenreg(models)
266
267 # Plot Coefficients
268 m2_coef_plot <- plot_coef(model_2_panel$model,
269                           model_name = "Model 2",
270                           sub = "bldegree + cycle(4)")
271 saveRDS(m2_coef_plot,
272         here::here("resources", "objects", "ergm", "m2_coef_plot.Rds"))
273 # ----- #
274 # Iteration 3 + MCMC Diagnostics + GOF
275 model_3_params <- bondora_net ~ edges + bldegree(1) + cycle(4) +
276
277 # See if there is gender effect
278 blfactor("bl_gender")
279
280 model_3 <- ergm::ergm(
281   model_3_params,
282   #constraints = ~ bd(minout = 0, maxout = 80),
283
284   control = ergm::control.ergm(
285     # Greater burn-in for cleaner result
286     # Cycles mixed with the rest of the terms requires more burnin samples
287     MCMC.burnin = 20000,
288     # Greater sample size for greater stability
289     MCMC.samplesize = 100000,
290     seed = 42,
291     # Reducing interval to reduce complexity between intervals
292     MCMC.interval = 1000,
293     # Only needed for convergence pvals to improve
294     MCMLE.maxit = 25,
295     # Smaller steps for stability
296     MCMLE.steplength = 0.25,
297     parallel = n_cores,
298     parallel.type = "PSOCK"
299   )
300 )
301
302 model_3_panel <- auto_ergm(model=model_3, mcmc=TRUE, name="ergm_m3")
303 snafun::stat_plot_gof(model_3_panel$gof)
304 model_3_panel$gof
305 models <- list(base_ergm, model_1, model_2, model_3)
306 texreg::screenreg(models)
307
308 # Plot Coefficients
309 m3_coef_plot <- plot_coef(model_3_panel$model,
310                           model_name = "Model 3",
311                           sub = "bldegree + cycle(4) + blfactor(gender)")
312 # Save Plot
313 saveRDS(m3_coef_plot,
314         here::here("resources", "objects", "ergm", "m3_coef_plot.Rds"))
315 # ----- #
316 # Iteration 4 + MCMC Diagnostics + GOF
317 model_4_params <- bondora_net ~ edges + bldegree(1) + cycle(4) +
318   blfactor("bl_gender") +
319
320 # See if higher ages make a difference
321 blcov("bl_age")
322
323 model_4 <- ergm::ergm(
324   model_4_params,
325   #constraints = ~ bd(minout = 0, maxout = 80),
326
327   control = ergm::control.ergm(
328     # Greater burn-in for cleaner result
329     MCMC.burnin = 20000,

```

```

330   # Greater sample size for greater stability
331   MCMC.samplesize = 100000,
332   seed = 42,
333   MCMC.interval = 1000,
334   # Only needed for convergence pvals to improve
335   MCMLE.maxit = 45,
336   # Smaller steps for stability
337   MCMLE.steplength = 0.25,
338   parallel = n_cores,
339   parallel.type = "PSOCK"
340 )
341 )
342
343 model_4_panel <- auto_ergm(model=model_4, mcmc=TRUE, name="ergm_m4")
344 model_4_panel$gof
345 snafun::stat_plot_gof(model_4_panel$gof)
346 models <- list(base_ergm, model_1, model_2, model_3, model_4)
347 texreg::screenreg(models)
348
349 # Plot Coefficients
350 m4_ergm_coef_plot <- plot_coef(model_4_panel$model,
351                                model_name = "Model 4",
352                                sub = "Unrestricted Model")
353
354 # Save Plot
355 saveRDS(m4_ergm_coef_plot,
356         here::here("resources", "objects", "ergm", "m4_coef_plot.Rds"))
357 # ----- #
358 # Collect Plots Together
359 gridExtra::grid.arrange(m1_coef_plot, m2_coef_plot,
360                          m3_coef_plot, m4_ergm_coef_plot,
361                          nrow = 2, ncol = 2)
362
363 # Collect Models in Table 1: Raw results
364 headers <- c("Base ERGM", "b1degree(1)", "cycle(4)", "Age", "Gender")
365 texreg::knitreg(models, custom.model.names = headers,
366                 digits = 3)
367
368 # Save the Table to use in Report
369 saveRDS(list(models, headers),
370         here::here("resources", "objects", "ergm", "ergm_table.Rds"))
371
372 # Padding
373 pad <- function(x, max) {
374   return(c(unname(x), rep("-", max-length(x))))
375 }
376
377 # Collect Models in Table 2: Log Odds Results
378 max_ergms <- length(model_4$coefficients)
379 df_prob <- data.frame(
380   Terms = names(model_4$coefficients),
381   "Base" = pad(round(lodds_to_prob(base_ergm$coefficients), 3), max_ergms),
382   "b1degree(1)" = pad(round(lodds_to_prob(model_1$coefficients), 3), max_ergms),
383   "cycle(4)" = pad(round(lodds_to_prob(model_2$coefficients), 3), max_ergms),
384   "Age" = pad(round(lodds_to_prob(model_3$coefficients), 3), max_ergms),
385   "Gender" = pad(round(lodds_to_prob(model_4$coefficients), 3), max_ergms)
386 )
387 knitr::kable(df_prob)
388
389 # Save Table to use in Report
390 saveRDS(df_prob,
391         here::here("resources", "objects", "ergm", "ergm_prob_tables.Rds"))
392 # ----- #

```

scripts/ergm_network_analysis.R

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B Technology Statement

During the preparation of this work, we used ChatGPT in order to generate select parts of the R script utilised to process the dataset. Specifically, the tool was used to transform the processed dataset into a format that igraph would accept as a network object. Additionally, some R functions written by the group were improved with ChatGPT's suggestions. No AI tool was utilised to independently write parts of the report. The following parts of the assignment were affected/generated by AI tool usage: **INTRODUCTION**; The tool was utilised to evaluate the validity of certain theoretical concepts and refine them. **DATASET**; the data described within this section was processed partly by some code drafted by ChatGPT and edited by the group. After using this tool/service, **Group 7** evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, **Group 7** takes full responsibility for the content of their work.